

The Theorem of Incompatible Truths

Why Any Sufficiently Complex Domain Necessarily Admits
Mutually Exclusive but Individually Valid Descriptions
That Cannot Be Unified

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Abstract

We investigate under what precise conditions a physical system necessarily admits multiple descriptions that are individually valid within their respective domains yet mutually incompatible when placed within a single formal framework. Rather than assuming any particular physical theory, we develop an independent formal apparatus built from five primitive notions—domain, description, validity, compatibility, and complexity—together with four axioms (Descriptive Completeness, Compression Imperative, Scale Coupling, and Non-Degeneracy). The central result is explicitly conditional: if a system satisfies all four axioms and exceeds a complexity threshold, incompatibility is provable. Our contribution is to identify these conditions precisely, to demonstrate that they are satisfied in concrete mathematical structures, and to show that relaxing any single condition eliminates incompatibility—at a cost we characterise exactly.

The investigation proceeds as a mathematical expedition with no predetermined conclusion. First, we attempt to establish the incompatibility thesis by five independent methods: an information-theoretic compression argument, a logical orthogonality argument, a category-theoretic non-existence-of-limit argument, a topological obstruction argument, and an effective-theory predictive-trade-off argument. These methods share a common structural root in the tension between compression and coverage but each reveals distinct formal consequences.

Second, we subject the thesis to sustained attack by attempting to prove its negation through four strategies: the trivial-domain construction, the hierarchy-of-approximations argument, the shared-ontology covering argument, and a systematic relaxation of each axiom. Third, we examine the possibility that the question of compatibility is itself undecidable from within the system.

To ground the abstract framework in concrete settings, we develop a fully formal instantiation in a finite lattice model where the primitives, axioms, descriptions, and incompatibility are exhibited without appeal to physical analogy. We additionally develop two computational toy models: a multi-scale agent simulation demonstrating that systems operating under locally incompatible rules outperform those constrained to a single unified rule set, and a forced-unification degradation model showing that imposing a single description across coupled subsystems with different natural dynamics leads to predictive collapse.

Every definition, proposition, and theorem is preceded and followed by thorough explanatory discussion. All computational code is provided in the appendices.

This paper constitutes the second article in a monograph. The first article established that any consistent physical system containing a proper-subsystem observer necessarily harbours ontological content inaccessible to all internal observers. The present article extends that result: not only is the full content inaccessible, but the accessible fragments are *mutually contradictory*. The connection between the two results is made precise. A brief remark indicates how both results relate to the Law of Imperative Uncertainty and the Law of Scale-Specific Principles, established in prior work, without developing that connection in full.

The paper reports what the mathematics yields, with minimal philosophical extrapolation beyond the formal results.

Contents

1	Introduction	6
1.1	The Central Question	6
1.2	Relation to the First Article	7
1.3	What This Paper Does and Does Not Do	7
1.4	Relation to Existing Work	8
1.5	Guide for the Reader	9
2	Primitive Notions	10
2.1	Domains	10
2.2	Descriptions	12
2.3	Validity	12
2.4	Compatibility	13
2.5	Complexity	14
3	The Four Axioms	16
4	Thesis I: The Information-Theoretic Compression Argument	19
4.1	The Compression–Coverage Trade-Off	20
4.2	The Incompatibility Theorem via Compression	21
4.3	Quantitative Bounds	22
4.4	Physical Illustration: Tunnelling vs. Electrostatics	23
5	Thesis II: The Logical Orthogonality Argument	24
5.1	Orthogonality of Axiom Systems	24
5.2	Alternative Proof via Model Theory	26
5.3	Physical Illustration: Reversibility vs. Irreversibility	26
6	Thesis III: The Category-Theoretic Non-Existence-of-Limit Argument	27
6.1	The Category of Descriptions	28
6.2	Physical Illustration: The Absence of a “Theory of Everything”	29
7	Thesis IV: The Topological Obstruction Argument	30
7.1	The Description Manifold	30
7.2	Cohomological Interpretation	31
8	Thesis V: The Predictive Trade-Off Argument	32
8.1	Predictive Accuracy and the Unification Penalty	32
8.2	The Kolmogorov Complexity Bound	34
9	Antithesis I: The Trivial-Domain Construction	35
9.1	The Claim	35
9.2	Does This Refute the Thesis?	35

10 Antithesis II: The Hierarchy-of-Approximations Argument	36
10.1 The Claim	36
10.2 Analysis: Does This Succeed?	36
11 Antithesis III: The Shared-Ontology Covering Argument	37
11.1 The Claim	38
11.2 Analysis	38
12 Antithesis IV: Systematic Axiom Relaxation	39
12.1 Systematic Relaxation	39
12.1.1 Relaxing Descriptive Completeness	39
12.1.2 Relaxing the Compression Imperative	39
12.1.3 Relaxing Scale Coupling	39
12.1.4 Relaxing Non-Degeneracy	39
12.2 Summary of Relaxation Results	40
13 The Independence Investigation	40
13.1 The Overlap Dilemma	41
13.2 Model-Theoretic Independence	41
14 Toy Models and Computational Demonstrations	42
14.1 Formal Instantiation: The Lattice Model	42
14.2 Toy Model 1: Multi-Scale Agent Simulation	44
14.3 Toy Model 2: Forced-Unification Degradation	46
15 Synthesis: What the Expedition Found	48
15.1 Summary of the Thesis Arguments	48
15.2 Summary of the Antithesis Attempts	49
15.3 Summary of the Undecidability Investigation	49
15.4 The Incompatibility Theorem	49
15.5 Interpretation	50
16 Discussion	51
16.1 Connection to the First Article	51
16.2 Connection to Prior Work	51
16.3 Implications for the “Theory of Everything” Programme	52
17 Conclusion	52
17.1 What Was Asked	52
17.2 What Was Found	52
17.3 What Was Not Found	53
17.4 Methodological Contribution	53
17.5 Closing Remark	53
References	54

A Computational Code	56
A.1 Toy Model 1: Dual-Regime Agent Simulation	56
A.2 Toy Model 2: Forced-Unification Degradation	57
B Peer Reviews and Revision History	58

1 Introduction

1.1 The Central Question

Consider a physical system that is sufficiently complex—a system with multiple organizational levels, each exhibiting stable, reproducible phenomena governed by identifiable regularities. At each level, competent observers construct descriptions: formal theories that compress observations into predictive frameworks. These descriptions work. They predict. They are tested and confirmed.

Now suppose two such descriptions, each valid within its own domain, are brought together. The question this paper investigates is:

Must there exist, in any sufficiently complex system, pairs of individually valid descriptions that logically contradict each other when placed within a single formal framework? And if so, is this contradiction eliminable—or is it a structural necessity?

This is not a question about the current state of human knowledge. It is not a claim that we have not yet found the right unifying language. It is a question about the formal structure of complexity itself: whether the very conditions that make a system complex enough to require multiple descriptions force those descriptions into mutual contradiction.

The question has obvious resonance with long-standing debates in the philosophy of science. The unity-of-science programme, from the Vienna Circle through Oppenheim and Putnam [Oppenheim and Putnam, 1958] to contemporary reductionism, assumes that all valid descriptions are in principle compatible—that apparent conflicts between theories at different levels reflect incomplete understanding, not structural impossibility. The disunity tradition, from Cartwright [Cartwright, 1999] and Dupré [Dupré, 1993] to the pluralist programmes of Mitchell [Mitchell, 2003] and Kellert, Longino, and Waters [Kellert et al., 2006], argues that science is irreducibly plural. But these arguments have remained largely philosophical. What is missing—and what this paper attempts to supply—is a *formal proof* that incompatibility is not merely observed but *mathematically necessary* above a complexity threshold.

The physical motivation is immediate and familiar. A particle with energy below a potential barrier *cannot* cross it according to classical electrostatics; the same particle *does* cross it according to quantum mechanics (tunnelling). Both descriptions are empirically confirmed within their respective domains. Placed in a single formal system, they yield a direct contradiction: “the particle crosses the barrier” and “the particle does not cross the barrier,” for the same particle, same barrier, same energy. This is not a failure of one theory. Both theories are correct where they apply.

The same pattern recurs throughout physics. Thermodynamics asserts the irreversibility of macroscopic processes; the microscopic dynamics underlying those same processes is time-reversible. At macroscopic scales, objects possess definite positions and momenta; at quantum scales, they exist in superposition. At cosmological scales, expansion dominates; at local astrophysical scales, gravitational binding dominates.

These are not exotic limit cases. They are the ordinary condition of physical reality across scales.

This paper formalises the conditions under which such contradictions are not contingent features of our current theories but *necessary consequences of complexity*. The result is explicitly conditional: it identifies four precise axioms and shows that their conjunction forces incompatibility. Our contribution is not to assert that physical systems satisfy these axioms—that is an empirical question—but to identify the exact conditions under which incompatibility is provable and to demonstrate that these conditions are satisfied in concrete, fully specified mathematical structures.

1.2 Relation to the First Article

This paper constitutes the second article in a monograph. The first article [Kriger, 2025a] established that any consistent physical system containing a proper-subsystem observer necessarily harbours ontological content inaccessible to all internal observers. The central result was a trichotomy: inaccessibility is provable when the observer is a proper subsystem, refutable when it is not, and independent of the consistency conditions when observer status is unspecified.

The present article extends that result in a specific direction. Article 1 showed that each observer’s description of the system is necessarily *incomplete*—there exist properties the observer cannot detect. The present article shows something stronger: that the descriptions constructed by different observers (or by the same observer operating at different scales) are not merely incomplete but *mutually contradictory*. Incompleteness alone does not imply incompatibility; two partial descriptions could agree on their shared content. The contribution of the present work is to prove that, above a complexity threshold, they cannot.

The logical relationship is as follows. Let $\mathcal{S}$ be a complex system, and let D_1, D_2 be valid descriptions of domains $\Omega_1, \Omega_2 \subseteq \mathcal{S}$. Article 1 established that each D_i is incomplete: there exist properties of Ω_i that D_i cannot represent. The present article establishes that when $\Omega_1 \cap \Omega_2 \neq \emptyset$ and the system’s complexity exceeds a threshold, the descriptions D_1 and D_2 *must* disagree on some statement about the overlap region. Incompleteness is a necessary condition for incompatibility (if a description were complete, it could in principle be checked for consistency with any other), but it is not sufficient. The additional structure that forces contradiction is the subject of the present work.

1.3 What This Paper Does and Does Not Do

We develop an independent formal apparatus built from five primitive notions—domain, description, validity, compatibility, and complexity—together with four axioms. From these definitions and axioms alone, without importing results from quantum mechanics, thermodynamics, or any other specific theory, we derive all subsequent results.

The investigation proceeds as a mathematical expedition with no predetermined conclusion. We pursue three lines of inquiry in sequence:

1. **The Incompatibility Thesis.** We attempt to prove, by five independent methods, that any system satisfying the four axioms with complexity above a threshold necessarily admits pairs of individually valid but mutually incompatible descriptions.
2. **The Compatibility Antithesis.** We then attempt, with equal seriousness, to prove the negation: that for any system satisfying the axioms, all valid descriptions can in principle be unified into a single consistent framework.
3. **The Undecidability Possibility.** Finally, we investigate whether an observer situated within the system can determine which of its available descriptions governs the overlap region where they contradict—and whether the question of compatibility is itself decidable from within.

The paper reports what the mathematics yields, with minimal extrapolation beyond the formal results. Where observations venture into interpretive territory, they are clearly flagged.

1.4 Relation to Existing Work

Several intellectual traditions bear on the present question. We engage each seriously but develop a framework independent of all.

Model theory and logical pluralism. The observation that different models of overlapping theories can disagree on shared language is standard in model theory. Our framework shares this ancestry. The specific contribution of the present work is threefold: (a) we identify the precise conditions (the four axioms) under which this disagreement is not merely possible but *necessary*; (b) we show that the conditions correspond to minimal requirements for “describing a complex system” (compression, completeness, coupling, diversity); and (c) we establish that the disagreement is non-eliminable, generic, and irresolvable from within. The individual techniques (compression arguments, basic model theory, category theory) are not novel; the contribution is their systematic application to a specific question about the structure of scientific descriptions, yielding a sharp conditional result.

Effective field theory and the renormalization group. The programme of effective field theory [Weinberg, 1979, Polchinski, 1984, Wilson and Kogut, 1974] provides the physical foundation: different energy scales are governed by different effective Lagrangians, and the renormalization group flow connecting them is generically irreversible. The Law of Scale-Specific Principles [Kriger, 2025b] formalises the methodological consequence. Our work differs in that we prove not merely that different scales *require* different descriptions (a methodological observation) but that those descriptions *must* contradict each other (a logical necessity).

Bohr’s complementarity. Bohr’s principle [Bohr, 1934] asserts that certain quantum phenomena require mutually exclusive experimental arrangements. Our framework generalises this beyond quantum mechanics to any system of sufficient complexity, and provides a formal proof rather than a philosophical principle.

Kuhnian incommensurability. Kuhn [Kuhn, 1962] argued that successive scientific paradigms are “incommensurable”—not merely different but untranslatable. Our work provides formal content to this intuition while differing in scope: we address not successive paradigms but *simultaneous* descriptions of the same system at different scales.

The disunity-of-science programme. Cartwright [Cartwright, 1999], Dupré [Dupré, 1993], and others have argued on philosophical grounds for the irreducible plurality of science. We provide mathematical proof where they provided argument.

Incompleteness and undecidability. Gödel’s incompleteness theorems [Gödel, 1931] and their extensions establish that sufficiently expressive formal systems are necessarily incomplete. Our result is different: we establish not that a single description is incomplete (which Article 1 already showed) but that multiple descriptions are *mutually inconsistent*. However, the diagonal and self-referential methods share a common ancestry with Gödel’s techniques.

The Law of Imperative Uncertainty. Kriger [Kriger, 2025c] proved that any system capable of sustaining nontrivial complexity must permit exceptions to its governing laws in the form of persistent uncertainty and probabilistic deviation. The present result is complementary: where the Law of Imperative Uncertainty establishes that a system must tolerate deviation from any single law, the Theorem of Incompatible Truths establishes that a system must tolerate contradiction between its laws. The connection between these results is noted in the Discussion (§16) but not developed in full, as it belongs to the third article of this monograph.

1.5 Guide for the Reader

The paper is constructed so that each section builds upon the previous ones, with definitions introduced precisely when needed. Section 2 introduces the five primitive notions. Section 3 develops the four axioms. The incompatibility thesis is then pursued through five proof methods in Sections 4–8. The compatibility antithesis is examined in Sections 9–12. The undecidability question is treated in Section 13. The toy models are developed in Section 14. Section 15 synthesises the results, Section 16 offers a brief discussion, and Section 17 concludes. Computational code for all toy models and figures appears in Appendix A. A placeholder for peer review and revision history is provided in Appendix B.

Each major step is accompanied by a boxed summary explaining its strategic purpose within the overall argument, and by a self-critique subsection that identifies potential weaknesses and either resolves them or flags them as open problems. Theorems are proved in full; where multiple proof methods are available, we present more than one.

Strategic Purpose

The introduction establishes three things. First, it poses the question in terms accessible to any scientifically literate reader, using concrete physical examples (tunnelling vs. electrostatics, irreversibility vs. reversibility) rather than abstract formalism. Second, it positions the paper relative to Article 1 and to existing

traditions without depending on any of them. Third, it sets the methodological ground rules: complete proofs, genuine self-critique, and no predetermined conclusion. This framing is necessary because the claim—that incompatibility is *provably necessary*, not merely observed—is strong enough to require careful grounding before the formal work begins.

Self-Critique and Resolution

Critique 1: The physical examples may prejudice the conclusion. By opening with tunnelling vs. electrostatics, the paper may create the impression that the result is obvious—“of course quantum and classical physics contradict.” This could lead readers to underestimate the strength of the formal claim.

Resolution. The examples are chosen for familiarity, not as evidence. The formal proof does not depend on any specific physical theory. The examples illustrate what the theorem predicts but do not constitute evidence for it. The distinction between illustration and proof is maintained throughout.

Critique 2: The relation to Article 1 may create a dependency. Readers who have not read Article 1 may feel unable to follow the argument.

Resolution. The present article is self-contained. All definitions and axioms are developed from scratch. Article 1 is referenced for context and for the specific result on inaccessibility that motivates the incompatibility question, but no proof in the present paper depends on Article 1’s machinery. A reader encountering this paper first will miss the broader monograph context but will follow the mathematics without difficulty.

Critique 3: The claim that incompatibility is “mathematically necessary” is very strong. Could it not be an artefact of the chosen axioms?

Resolution. This is the correct question, and it is addressed head-on. The antithesis sections (§9–§12) systematically relax each axiom and examine whether incompatibility survives. The result is that incompatibility is tightly coupled to the axioms: it holds when all four are satisfied and can fail when any one is dropped. Whether the axioms themselves are “correct” is a question the paper addresses by showing that they capture minimal and physically natural conditions. The reader is invited to reject any axiom and observe the consequence.

2 Primitive Notions

This section introduces the five foundational concepts upon which the entire paper rests: domain, description, validity, compatibility, and complexity. Each is defined with the minimum structure necessary for the subsequent proofs, and each definition is accompanied by discussion of what it captures and what it deliberately leaves open.

2.1 Domains

We begin with the notion of a region of reality that can be observed and described.

Definition 2.1 (Domain). A *domain* is a quadruple $\Omega = (\Sigma_\Omega, D_\Omega, \mathbf{t}, \mathcal{O}_\Omega)$ where:

1. Σ_Ω is a non-empty set, called the *state space* of the domain;
2. $\mathbf{t}$ is a non-empty totally ordered commutative semigroup (the time structure);
3. $D_\Omega : \Sigma_\Omega \times \mathbf{t} \rightarrow \Sigma_\Omega$ is the *dynamics*, satisfying the semigroup property $D_\Omega(D_\Omega(\sigma, \tau_1), \tau_2) = D_\Omega(\sigma, \tau_1 + \tau_2)$;
4. $\mathcal{O}_\Omega \subseteq \mathcal{F}(\Sigma_\Omega, \mathbb{R})$ is a non-empty collection of real-valued functions on Σ_Ω , called the *stable observables* of the domain.

A domain is a portion of a physical system viewed at a particular scale or resolution. The state space Σ_Ω gathers all configurations accessible within that domain. The dynamics D_Ω governs how states evolve. The stable observables $\mathcal{O}_\Omega$ are the measurable quantities that exhibit reproducible, predictable behaviour—the regularities that any description of the domain must account for.

The inclusion of stable observables in the definition is deliberate. A domain is not merely a set of states with dynamics; it is a set of states with dynamics *and* identifiable regularities. This distinction matters because the incompatibility thesis concerns descriptions that predict these regularities. A domain without stable observables has nothing to describe and nothing to contradict.

Remark 2.2 (Relation to Article 1). The domain definition inherits the state-space and dynamics structure from Article 1’s physical system definition (Definition 2.1 of [Kriger, 2025a]), adding the stable-observable component. This addition is necessary because the present paper concerns the content of descriptions, not merely the structure of the underlying system.

Definition 2.3 (Domain Overlap). Two domains $\Omega_1 = (\Sigma_1, D_1, \mathbf{t}, \mathcal{O}_1)$ and $\Omega_2 = (\Sigma_2, D_2, \mathbf{t}, \mathcal{O}_2)$ *overlap* if there exist embedding maps $\iota_1 : \Sigma_\cap \hookrightarrow \Sigma_1$ and $\iota_2 : \Sigma_\cap \hookrightarrow \Sigma_2$ from a non-empty set $\Sigma_\cap$ (the *overlap state space*) such that:

1. ι_1 and ι_2 are injective;
2. The dynamics are compatible on the overlap: for every $\sigma \in \Sigma_\cap$ and $\tau \in \mathbf{t}$, the images $\iota_1(D_\cap(\sigma, \tau))$ and $D_1(\iota_1(\sigma), \tau)$ agree, and similarly for ι_2 ;
3. There exist observables $f_1 \in \mathcal{O}_1$ and $f_2 \in \mathcal{O}_2$ that, when restricted to the overlap via the embeddings, measure the same physical quantity: $f_1 \circ \iota_1 = f_2 \circ \iota_2$ as functions on $\Sigma_\cap$.

We write $\Omega_1 \cap \Omega_2 \neq \emptyset$ and call the triple $(\Sigma_\cap, \iota_1, \iota_2)$ the *overlap structure*.

The overlap structure is the formal expression of the physical situation where two domains share phenomena. Condition (3) is essential: for two descriptions to potentially contradict each other, they must make predictions about the same observable quantity in the same region of state space. Without shared observables, the descriptions are simply about different things and cannot conflict.

The physical example is immediate. The quantum-mechanical domain and the classical electrostatic domain overlap in the regime of a particle encountering a potential barrier. Both domains contain the observable “particle position relative to the barrier.” It is precisely this shared observable that generates the contradiction: quantum mechanics predicts a nonzero probability of finding the particle beyond the barrier; classical electrostatics predicts zero probability.

2.2 Descriptions

A description is a formal theory that makes predictions about a domain’s observables.

Definition 2.4 (Description). A *description* of a domain Ω is a triple $\mathcal{D} = (\mathcal{L}, \mathcal{A}, \vdash)$ where:

1. $\mathcal{L}$ is a formal language: a set of well-formed sentences, including sentences of the form $\ulcorner f(\sigma) \in S \urcorner$ for observables $f \in \mathcal{O}_\Omega$ and measurable sets $S \subseteq \mathbb{R}$;
2. $\mathcal{A} \subset \mathcal{L}$ is a finite set of axioms;
3. $\vdash$ is a consequence relation (derivability) satisfying the standard structural rules (reflexivity, monotonicity, transitivity/cut).

The *deductive closure* of $\mathcal{D}$ is $\text{Th}(\mathcal{D}) = \{\varphi \in \mathcal{L} \mid \mathcal{A} \vdash \varphi\}$.

A description is, in the language of mathematical logic, a finitely axiomatised theory in a formal language rich enough to talk about the domain’s observables. The requirement that $\mathcal{A}$ be finite reflects the physical constraint that any usable theory must be finitely specifiable—an observer with bounded resources cannot store or manipulate an infinite axiom set.

The formal language $\mathcal{L}$ must be expressive enough to make predictions about observables. The sentence $\ulcorner f(\sigma) \in S \urcorner$ asserts that observable f , evaluated at state σ , takes a value in the set S . For instance, “the particle’s position is beyond the barrier” is a sentence of this form.

Definition 2.5 (Consistency of a Description). A description $\mathcal{D} = (\mathcal{L}, \mathcal{A}, \vdash)$ is *consistent* if there exists no sentence $\varphi \in \mathcal{L}$ such that both $\mathcal{A} \vdash \varphi$ and $\mathcal{A} \vdash \neg\varphi$.

We require each individual description to be internally consistent. The question of the paper is whether *pairs* of individually consistent descriptions can be jointly consistent.

2.3 Validity

A description is valid if its predictions match what is actually observed.

Definition 2.6 (Validity). A description $\mathcal{D}$ of a domain Ω is *valid* for Ω if:

1. $\mathcal{D}$ is consistent;

2. For every stable observable $f \in \mathcal{O}_\Omega$, every state $\sigma \in \Sigma_\Omega$, and every prediction $\ulcorner f(\sigma) \in S \urcorner \in \text{Th}(\mathcal{D})$, the prediction is empirically correct: $f(\sigma) \in S$ holds in the domain.

We write $\mathcal{D} \models \Omega$ to denote that $\mathcal{D}$ is valid for Ω .

Validity is the conjunction of internal consistency and external correctness. A valid description does not contradict itself and does not contradict observation. This is a strong condition: it requires that *every* prediction the description makes about stable observables is correct. A description that makes even one false prediction about a stable observable is not valid.

Note what validity does *not* require: it does not require completeness. A valid description may remain silent about many observables. It must not be wrong about anything it does predict, but it need not predict everything. This distinction is important: the Descriptive Completeness axiom (§3) will constrain how much silence is permitted.

2.4 Compatibility

Two descriptions are compatible if they can coexist in a single consistent framework.

Definition 2.7 (Compatibility). Two descriptions $\mathcal{D}_1 = (\mathcal{L}_1, \mathcal{A}_1, \vdash_1)$ and $\mathcal{D}_2 = (\mathcal{L}_2, \mathcal{A}_2, \vdash_2)$ of overlapping domains Ω_1, Ω_2 are *compatible* if there exists a description $\mathcal{D}_U = (\mathcal{L}_U, \mathcal{A}_U, \vdash_U)$ such that:

1. $\mathcal{L}_U \supseteq \mathcal{L}_1 \cup \mathcal{L}_2$ (the unified language contains both languages);
2. $\mathcal{A}_U$ is consistent;
3. $\text{Th}(\mathcal{D}_U) \supseteq \text{Th}(\mathcal{D}_1) \cup \text{Th}(\mathcal{D}_2)$ restricted to their respective domains.

Two descriptions that are not compatible are *incompatible*.

Compatibility asks: can the two descriptions be embedded in a single consistent super-theory that preserves all predictions of both? If yes, they are compatible. If no—if every attempt to combine them into a single consistent framework requires abandoning some predictions of one or both—they are incompatible.

The condition is demanding. It does not merely ask whether the axioms $\mathcal{A}_1 \cup \mathcal{A}_2$ are jointly consistent; it asks whether all their *consequences* restricted to the respective domains can coexist. Two axiom sets might be jointly consistent while their deductive closures conflict on the overlap.

Remark 2.8 (Three levels of incompatibility). The definition captures what we may call *logical incompatibility*. The paper will also address two further levels:

- *Linguistic incompatibility*: no single formal language can express both descriptions without loss of content (the descriptions are semantically incommensurable);
- *Predictive incompatibility*: any unified description that attempts to cover both domains suffers predictive degradation in at least one.

These three levels correspond to the three planes identified in our programme. The logical level is formalised in the definition above; the linguistic and predictive levels are developed in the thesis sections.

2.5 Complexity

The incompatibility thesis is not a universal claim about all systems. It applies above a *complexity threshold*. We define complexity through multiple measures simultaneously, ensuring that the threshold result is robust and not an artefact of any single formalism.

Definition 2.9 (Complexity Measures). Let $\mathcal{S}$ be a physical system containing domains $\Omega_1, \dots, \Omega_n$. We define the following complexity measures:

1. **Kolmogorov complexity** $K(\mathcal{S})$: the length of the shortest programme (in a fixed universal Turing machine) that produces a complete specification of $\mathcal{S}$'s state space, dynamics, and observables.
2. **Effective complexity** $C_{\text{eff}}(\mathcal{S})$ (following Gell-Mann and Lloyd [Gell-Mann and Lloyd, 1996]): the length of the shortest description of the system's *regularities*, excluding random or incompressible components.
3. **Logical depth** $d(\mathcal{S})$ (following Bennett [Bennett, 1988]): the computational time required to produce $\mathcal{S}$'s specification from its shortest programme.
4. **Descriptive diversity** $\delta(\mathcal{S})$: the number of distinguishable domains Ω_i within $\mathcal{S}$ that require descriptions with non-isomorphic axiom sets. Formally, $\delta(\mathcal{S}) = |\{[\mathcal{D}_i] : \mathcal{D}_i \models \Omega_i, [\cdot]$ denotes logical equivalence $\}|$.

Each measure captures a different aspect of what makes a system “complex enough” for the incompatibility thesis to apply. Kolmogorov complexity measures the raw information content. Effective complexity measures the structured, non-random information. Logical depth measures the irreducibility of the computational process that generates the system's regularities. Descriptive diversity measures how many genuinely different theories are needed.

The fourth measure—descriptive diversity—is original to this work and is the most directly relevant. A system with $\delta(\mathcal{S}) = 1$ can be described by a single theory (possibly with parameters varying across domains). A system with $\delta(\mathcal{S}) \geq 2$ requires at least two logically non-equivalent theories. The incompatibility thesis asserts that when $\delta(\mathcal{S}) \geq 2$ and the domains overlap, the required theories must contradict each other.

Definition 2.10 (Complexity Threshold). The system $\mathcal{S}$ *exceeds the complexity threshold* if:

1. $\delta(\mathcal{S}) \geq 2$: at least two non-isomorphic descriptions are required;
2. $C_{\text{eff}}(\mathcal{S}) > \max_i C_{\text{eff}}(\Omega_i)$: the system's regularities are not reducible to those of any single domain.

The remaining measures ($K(\mathcal{S})$ and $d(\mathcal{S})$) provide independent corroboration but are not invoked in the proofs. The essential condition is $\delta(\mathcal{S}) \geq 2$ together with the four axioms; condition (2) ensures that the diversity is not trivial (arising from superficial relabelling).

Remark 2.11 (Honesty about the threshold). Condition (1) is close to presupposing the setup for the incompatibility result, since it asserts that genuinely different descriptions exist. We are explicit about this: the complexity threshold is not an independently motivated measure but a *precisely identified precondition*. The theorem says: “given a system where these conditions hold, incompatibility follows.” Whether any particular physical system satisfies these conditions is an empirical question that the theorem does not answer. The formal instantiation in §14.1 demonstrates that concrete systems satisfying all conditions exist.

Strategic Purpose

Defining five primitive notions from scratch ensures that the paper is self-contained and does not inherit assumptions from any particular physical theory. The domain definition captures “a region of reality at a particular scale”; the description definition captures “a formal theory that predicts observables”; validity ensures descriptions are correct where they apply; compatibility formalises what “unification” would mean; and complexity, measured by four independent metrics, ensures that the threshold result is definition-independent. The overlap structure is the critical innovation: it provides the formal arena where two descriptions can confront each other, and it will be the site of every subsequent contradiction.

Self-Critique and Resolution

Critique 1: The domain definition assumes a global time structure. As noted in Article 1 (Remark 2.2), a totally ordered time structure excludes relativistic systems.

Resolution. The limitation is inherited from Article 1 for consistency across the monograph. The results apply to systems with a global time parameter—including non-relativistic quantum mechanics, classical mechanics, and discrete-time systems. Extension to partially ordered time structures is noted as future work.

Critique 2: “Stable observable” is undefined. The definition requires $\mathcal{O}_\Omega$ to be a collection of “stable observables” but does not define stability.

Resolution. Stability is understood operationally: an observable is stable if repeated measurements under the same conditions yield reproducible results within the domain. A formal definition would require specifying a topology on the space of observables and a notion of convergence, which would add machinery without strengthening the results. The theorems require only that stable observables exist and are shared across overlapping domains; the precise criterion for stability is not invoked.

Critique 3: The compatibility definition is very strong. Requiring the unified theory to preserve *all* consequences of both descriptions may be too demanding.

Resolution. This is deliberate. A weaker definition (e.g., preserving only the axioms, not all consequences) would make compatibility easier to achieve and the in-

compatibility thesis harder to prove. By choosing the strongest reasonable definition of compatibility, we make the incompatibility result as robust as possible. The antithesis sections explore whether weakening the definition allows compatibility, and identify the precise cost of each weakening.

3 The Four Axioms

The incompatibility thesis does not follow from the primitive notions alone. Four additional structural conditions are needed, each capturing a minimal and physically natural constraint on how descriptions relate to the systems they describe. We state each axiom, explain its content, and show why it is necessary by exhibiting what fails without it.

Axiom 1 (Descriptive Completeness). If a description $\mathcal{D}$ is valid for a domain Ω , then for every stable observable $f \in \mathcal{O}_\Omega$, there exists a non-trivial prediction: a sentence $\lceil f(\sigma) \in S \rceil \in \text{Th}(\mathcal{D})$ for some proper subset $S \subsetneq \mathbb{R}$ and some state $\sigma \in \Sigma_\Omega$. A valid description cannot remain entirely silent about any stable observable within its domain.

Descriptive Completeness does not require omniscience. It does not demand that a description predict the value of every observable at every state. It requires only that the description say *something non-trivial* about each stable observable—that it constrain the observable’s possible values to a proper subset of $\mathbb{R}$ for at least one state.

This axiom is necessary because without it, two descriptions could trivially avoid contradiction by dividing the observables between them, each remaining silent about the other’s subject matter. Such descriptions would be “compatible” in a vacuous sense: they never disagree because they never talk about the same thing. Descriptive Completeness prevents this evasion by requiring that every valid description engages with all the stable observables of its domain.

The physical motivation is immediate. A theory of classical electrostatics that remained silent about electric fields would not be a theory of electrostatics. A theory of quantum mechanics that made no predictions about position measurements would not be quantum mechanics. Stable observables define the domain; a description of the domain must address them.

Axiom 2 (Compression Imperative). A valid description $\mathcal{D} = (\mathcal{L}, \mathcal{A}, \vdash)$ of a domain Ω must compress: $K(\mathcal{D}) < K(\text{raw data of } \Omega)$, where K denotes Kolmogorov complexity. For finite systems, this takes the explicit form

$$|\mathcal{A}| + |\mathcal{L}_{\text{syntax}}| < |\mathcal{O}_\Omega| \cdot |\Sigma_\Omega|,$$

where $|\mathcal{A}|$ is the total length of the axioms (in some fixed encoding), $|\mathcal{L}_{\text{syntax}}|$ is the specification length of the language, and $|\mathcal{O}_\Omega| \cdot |\Sigma_\Omega|$ is the size of a brute-force enumeration. For continuous state spaces and observable sets, the Kolmogorov formulation applies directly: the algorithmic complexity of the description must be strictly less than the algorithmic complexity of a lookup table specifying all observable values to any desired precision.

The Compression Imperative states that a description must be shorter than a raw tabulation of the data it accounts for. A “theory” that simply lists every measurement result—“at state σ_1 , observable f_1 has value 3.7; at state σ_2 , observable f_1 has value 2.1; ...”—is not a theory at all. It is a database. A genuine description compresses: it captures the regularities in fewer symbols than it would take to state all the individual facts.

This axiom is necessary because without it, a trivial “unified description” could always be constructed: simply concatenate all the predictions of $\mathcal{D}_1$ and $\mathcal{D}_2$ into a single list. Such a concatenation is formally consistent (assuming $\mathcal{D}_1$ and $\mathcal{D}_2$ are individually consistent and their predictions happen not to explicitly contradict on the overlap), but it is not a theory—it does not compress, does not explain, and does not predict anything beyond what it explicitly lists. The Compression Imperative excludes such pseudo-unifications.

The physical motivation is foundational to science itself. Newton’s laws compress the motions of all gravitationally interacting bodies into three short statements. Maxwell’s equations compress all of classical electrodynamics into four. The power of a theory lies precisely in its compression ratio: the degree to which it says more than it stores.

Axiom 3 (Scale Coupling). In systems above the complexity threshold $C_{\text{threshold}}$, there exist at least two domains Ω_1, Ω_2 with $\Omega_1 \cap \Omega_2 \neq \emptyset$: the domains overlap, sharing at least one stable observable in the overlap region.

Scale Coupling asserts that the domains of a complex system are not isolated islands. They share boundaries; their phenomena interpenetrate; there exist observables that belong to both domains simultaneously. This is the condition that creates the arena for potential contradiction.

Without Scale Coupling, the incompatibility thesis is trivially false: if no two domains share observables, their descriptions operate in disjoint universes and cannot conflict. The physical universe is not like this. Quantum effects manifest in macroscopic phenomena (superconductivity, laser operation, tunnelling in semiconductor devices). Thermodynamic behaviour constrains microscopic dynamics (the fluctuation-dissipation theorem). Cosmological expansion affects local gravitational dynamics (the Hubble flow). The domains of physics are coupled, and their coupling is the source of the deepest conflicts between their descriptions.

Axiom 4 (Non-Degeneracy). The system $\mathcal{S}$ contains at least two domains Ω_1, Ω_2 whose valid descriptions $\mathcal{D}_1, \mathcal{D}_2$ have non-isomorphic deductive closures: $\text{Th}(\mathcal{D}_1) \not\cong \text{Th}(\mathcal{D}_2)$. Equivalently, the descriptive diversity $\delta(\mathcal{S}) \geq 2$.

Non-Degeneracy excludes the trivial case where a single description (possibly with different parameter values) covers all domains. If $\mathcal{D}_1$ and $\mathcal{D}_2$ are logically equivalent—if they make the same predictions, merely expressed in different notation—then “incompatibility” is meaningless.

The axiom requires that the two descriptions are *genuinely different*: they make different structural assumptions, employ different concepts, and derive their predictions

through different logical paths. Classical electrostatics and quantum mechanics are genuinely different in this sense. Newtonian mechanics and general relativity are genuinely different. Two parametrisations of the same effective field theory with different coupling constants are *not* genuinely different (their deductive closures are isomorphic under parameter relabelling).

Proposition 3.1 (Independence of Axioms). *The four axioms are logically independent: no proper subset implies the remaining axiom(s).*

Proof. We exhibit, for each axiom, a scenario satisfying the other three but violating that one.

(i) *Descriptive Completeness violated, others satisfied.* Consider a system with two overlapping domains, each admitting a genuinely different valid description, where the descriptions are compressive—but each description remains silent about the shared observable in the overlap region, making predictions only about observables unique to its own domain. The descriptions cannot contradict because they never address the same observable. Scale Coupling, Compression Imperative, and Non-Degeneracy all hold.

(ii) *Compression Imperative violated, others satisfied.* Consider a system with two overlapping domains, each admitting a genuinely different “description” that is a brute-force enumeration of all observable values. Both “descriptions” address all observables (Descriptive Completeness), the domains overlap (Scale Coupling), and the enumerations are non-isomorphic (Non-Degeneracy). The two lists can be trivially merged into a single consistent list. No compression means no structural commitment, so no contradiction arises.

(iii) *Scale Coupling violated, others satisfied.* Consider a system with two non-overlapping domains, each with a compressive, complete, genuinely different valid description. Since the domains share no observables, the descriptions cannot conflict regardless of their content.

(iv) *Non-Degeneracy violated, others satisfied.* Consider a system with two overlapping domains, both admitting the same (up to isomorphism) compressive, complete description. The descriptions agree on everything because they are the same theory. Scale Coupling holds, Compression and Completeness hold, but there is no genuine diversity. $\square$

Strategic Purpose

The four axioms serve a dual purpose. First, they make precise the conditions under which the incompatibility thesis holds, transforming a vague intuition (“complex systems have contradictory descriptions”) into a sharp conditional statement. Second, they provide four “knobs” for the antithesis: when we later attempt to refute the incompatibility thesis, we will systematically relax each axiom and examine the consequences. This parallels the methodological structure of Article 1, where four consistency conditions served the same role. The independence proof ensures that the axioms are genuinely separate constraints, not redundant restatements of a single idea.

Self-Critique and Resolution

Critique 1: The Compression Imperative formalises an intuition about “genuine theories” that may be contentious. Some philosophers of science reject the identification of explanation with compression (e.g., Woodward’s interventionist account [Woodward, 2003]).

Resolution. The Compression Imperative is not a claim about what constitutes explanation in general. It is a minimal condition on what constitutes a *formal description* in the sense of this paper: a finite specification shorter than a raw data table. This is uncontroversial in information theory and in the practice of theoretical physics, where “theory” is understood as compression of empirical regularities into laws. Readers who reject the Compression Imperative are invited to examine Antithesis IV (§12), where we show that dropping it permits trivial unification—at the cost of the unified “theory” having no explanatory or predictive power beyond what was already known.

Critique 2: Scale Coupling is assumed, not proved. Why should domains overlap? Could a complex system have perfectly isolated domains?

Resolution. Scale Coupling is an axiom precisely because it captures an empirical fact about physical systems that cannot be derived from purely formal considerations. Complex systems in nature exhibit coupled domains because the physical processes at different scales interact (e.g., quantum effects influence macroscopic behaviour via tunnelling, entanglement, and decoherence). A system with perfectly isolated domains would satisfy a very specific structural condition (complete decoupling across scales) that is non-generic in the space of dynamical systems. The axiom asserts that we are considering the generic case.

Critique 3: Descriptive Completeness may be too strong. Requiring a description to say something about every stable observable precludes theories that deliberately restrict their scope.

Resolution. Descriptive Completeness applies *within the domain*. A theory of electrostatics is not required to predict nuclear reactions; it is required to predict all stable electromagnetic observables within its domain. The scope restriction happens at the level of domain definition, not at the level of the axiom. A description’s domain is chosen to match its scope, and within that domain, Completeness is a natural requirement.

4 Thesis I: The Information-Theoretic Compression Argument

We now present the first of five arguments for the incompatibility thesis. This argument exploits the tension between the Compression Imperative and Descriptive Completeness: any valid description must compress (say more than it stores), but compression requires choosing which features to treat as fundamental and which to derive or discard. Different domains, with different dominant regularities, force different compression choices—and these choices, when confronted on the overlap, produce contradictions.

4.1 The Compression–Coverage Trade-Off

The core idea is formalised as follows.

Definition 4.1 (Compression Scheme). Given a domain $\Omega = (\Sigma_\Omega, D_\Omega, \mathbf{t}, \mathcal{O}_\Omega)$, a *compression scheme* for Ω is a pair (P, R) where:

1. $P \subseteq \mathcal{O}_\Omega$ is a set of *primitive observables* (those treated as fundamental);
2. $R : \mathcal{O}_\Omega \setminus P \rightarrow \text{Expr}(P)$ is a *reduction map* that expresses each non-primitive observable as a function of the primitives;
3. *Functional independence of primitives*: for each $f \in P$, there exist states $\sigma_1, \sigma_2 \in \Sigma_\Omega$ such that $f(\sigma_1) \neq f(\sigma_2)$ while $g(\sigma_1) = g(\sigma_2)$ for all $g \in P \setminus \{f\}$. That is, each primitive carries information not reducible to the others.

The *compression cost* is $|P|$ (the number of primitives). The *coverage* is $|\mathcal{O}_\Omega|$ (the number of observables accounted for). The *compression ratio* is $\rho = |\mathcal{O}_\Omega|/|P|$.

The functional independence condition (3) is essential and addresses a subtle point. A compression scheme is not merely a labelling of some observables as “primitive.” It is a commitment that the primitives are *genuinely independent*—each carries information not present in the others. Without this condition, two descriptions could share the same primitive set but differ only in labelling, and the subsequent incompatibility argument would fail. The condition ensures that calling an observable “primitive” has real content: it means the observable varies freely, unconstrained by the other primitives.

Every valid description implicitly defines a compression scheme: its axioms select certain observables as primitive (those appearing directly in the axioms), and its theorems express all other observables in terms of the primitives. The Compression Imperative requires $\rho > 1$: the description must cover more observables than it takes as primitive.

The following lemma establishes that compression is not free—it requires structural commitments that constrain what the description can say.

Lemma 4.2 (Compression Forces Structural Commitment). *Let $\mathcal{D}$ be a valid, compressive description of domain Ω with compression scheme (P, R) . Then $\mathcal{D}$ entails specific functional relationships among the observables: for every $f \in \mathcal{O}_\Omega \setminus P$, the description $\mathcal{D}$ asserts that $f = R(f)$, i.e., that f is determined by the primitives via the reduction map R .*

Proof. By definition, $R(f) \in \text{Expr}(P)$ is an expression in terms of the primitives. The validity of $\mathcal{D}$ requires that $R(f)$ agrees with f on all states in Σ_Ω : for every $\sigma \in \Sigma_\Omega$, $f(\sigma) = R(f)(\sigma)$. This agreement is a non-trivial assertion—it constrains the relationships among observables. Without it, the compression scheme would not be valid (the reduced expression would disagree with the observable). $\square$

This lemma articulates a point that is intuitively clear but formally important: compression is not just abbreviation. It is the assertion that certain relationships hold—that certain observables are derivable from others. These assertions are structural commitments, and they can conflict.

4.2 The Incompatibility Theorem via Compression

Theorem 4.3 (Compression Incompatibility). *Let $\mathcal{S}$ be a system satisfying Axioms 1–4 with complexity exceeding $C_{\text{threshold}}$. Let Ω_1, Ω_2 be two overlapping domains with valid descriptions $\mathcal{D}_1, \mathcal{D}_2$ having compression schemes (P_1, R_1) and (P_2, R_2) respectively. If $P_1 \neq P_2$ (the two descriptions choose different primitive observables) and the domains share at least one stable observable $f^* \in \mathcal{O}_1 \cap \mathcal{O}_2$ that is primitive in one description and derived in the other, then $\mathcal{D}_1$ and $\mathcal{D}_2$ are incompatible.*

Before proving this, we unpack the hypothesis. The condition $P_1 \neq P_2$ says that the two descriptions treat different observables as fundamental. The condition that a shared observable f^* is primitive in one and derived in the other is the critical one: it means that $\mathcal{D}_1$ takes f^* as a basic fact of nature, while $\mathcal{D}_2$ asserts that f^* is a consequence of other, more fundamental quantities. This asymmetry generates the contradiction.

Proof. We construct an explicit contradictory statement.

Since $f^* \in P_1$ (primitive in $\mathcal{D}_1$), the description $\mathcal{D}_1$ treats f^* as an independent observable: its value at any state is not constrained by the values of other observables. In particular, $\mathcal{D}_1$ admits states $\sigma \in \Sigma_\cap$ (in the overlap region) where $f^*(\sigma)$ is unconstrained by the other primitives of $\mathcal{D}_1$.

Since $f^* \notin P_2$ (derived in $\mathcal{D}_2$), the description $\mathcal{D}_2$ asserts $f^* = R_2(f^*)$, where $R_2(f^*)$ is an expression in terms of the primitives P_2 . This means $\mathcal{D}_2$ asserts that f^* is *determined* by the primitives of P_2 : its value at any state is a specific function of the other observables.

Now consider the overlap region $\Sigma_\cap$. On this region, both descriptions must make predictions (by Axiom 1 and Axiom 3). The descriptions entail:

$$\mathcal{D}_1 \vdash \exists \sigma_1, \sigma_2 \in \Sigma_\cap : g(\sigma_1) = g(\sigma_2) \wedge f^*(\sigma_1) \neq f^*(\sigma_2) \quad (1)$$

$$\mathcal{D}_2 \vdash \forall \sigma_1, \sigma_2 \in \Sigma_\cap : [\bigwedge_{h \in P_2} h(\sigma_1) = h(\sigma_2)] \Rightarrow f^*(\sigma_1) = f^*(\sigma_2) \quad (2)$$

where $g \in P_2 \cap \mathcal{O}_\cap$ is a shared observable that is primitive in $\mathcal{D}_2$.

Statement (1) asserts that f^* can vary independently of g . Statement (2) asserts that f^* is determined by the P_2 -primitives (including g).

Any unified description $\mathcal{D}_U$ containing both $\mathcal{D}_1$ and $\mathcal{D}_2$ would derive both statements. For the contradiction to materialise, we need the states σ_1, σ_2 from (1) to also agree on all P_2 -primitives other than those linked to f^* .

We show this is necessarily the case when $\Sigma_\cap$ is finite (the case most directly supported by the framework's definitions; the continuous case is treated via the topological argument of §7).

By the functional independence condition (Definition 4.1, part 3), since $f^* \in P_1$, there exist $\sigma_1, \sigma_2 \in \Sigma_\cap$ with $f^*(\sigma_1) \neq f^*(\sigma_2)$ while all other primitives of P_1 agree: $h(\sigma_1) = h(\sigma_2)$ for all $h \in P_1 \setminus \{f^*\}$. This is a direct consequence of the definition, not an additional assumption.

Now consider the P_2 -primitives evaluated on σ_1, σ_2 . Since the P_2 -primitives are a different set from the P_1 -primitives (by hypothesis $P_1 \neq P_2$), and both are finite, the P_2 -primitive values on σ_1, σ_2 are determined by the system, not by $\mathcal{D}_1$'s axioms. There are two sub-cases:

Sub-case (a): All P_2 -primitives agree on σ_1, σ_2 : $h(\sigma_1) = h(\sigma_2)$ for all $h \in P_2$. Then $\mathcal{D}_2$ predicts $f^*(\sigma_1) = R_2(f^*)(\sigma_1) = R_2(f^*)(\sigma_2) = f^*(\sigma_2)$, since $R_2(f^*)$ is a function of the P_2 -primitives. But we chose σ_1, σ_2 so that $f^*(\sigma_1) \neq f^*(\sigma_2)$. This is a direct contradiction.

Sub-case (b): Some P_2 -primitive disagrees on σ_1, σ_2 . Then $\mathcal{D}_1$ and $\mathcal{D}_2$ disagree about the *dependencies*: $\mathcal{D}_1$ asserts that f^* varies while $P_1 \setminus \{f^*\}$ is constant (independence), while $\mathcal{D}_2$ asserts that whenever the P_2 -primitives vary, f^* varies only as prescribed by $R_2(f^*)$ (dependence). The sentence $\varphi \equiv \lceil \exists \sigma_1, \sigma_2 : [\bigwedge_{h \in P_2} h(\sigma_1) = h(\sigma_2)] \wedge f^*(\sigma_1) \neq f^*(\sigma_2) \rceil$ is derivable from $\mathcal{D}_1$ (if sub-case (a) applies to a different pair of witness states, which exists by iterating the independence condition) and its negation is derivable from $\mathcal{D}_2$ (by the definition of the reduction map).

In either sub-case, the unified description $\mathcal{D}_U$ is inconsistent. $\square$

Key Insight

The compression argument reveals the source of incompatibility with particular clarity: it is the *choice of what is fundamental*. Description $\mathcal{D}_1$ says “ f^* is a basic fact.” Description $\mathcal{D}_2$ says “ f^* is a consequence of deeper facts.” Both are correct within their domains. But “ f^* is independent” and “ f^* is determined” are logically incompatible assertions. No unified framework can affirm both.

4.3 Quantitative Bounds

The compression argument yields quantitative bounds on the degree of incompatibility.

Proposition 4.4 (Incompatibility Measure). *Let $\mathcal{D}_1, \mathcal{D}_2$ be valid descriptions of overlapping domains with compression schemes (P_1, R_1) and (P_2, R_2) . Define the asymmetric overlap as*

$$\alpha_{12} = |\{f \in \mathcal{O}_\cap : f \in P_1 \setminus P_2\}|, \quad \alpha_{21} = |\{f \in \mathcal{O}_\cap : f \in P_2 \setminus P_1\}|,$$

where $\mathcal{O}_\cap = \mathcal{O}_1 \cap \mathcal{O}_2$ is the set of shared observables. The incompatibility degree is

$$\iota(\mathcal{D}_1, \mathcal{D}_2) = \min_{(P_1, R_1), (P_2, R_2)} (\alpha_{12} + \alpha_{21}),$$

where the minimum is taken over all valid compression schemes for $\mathcal{D}_1$ and $\mathcal{D}_2$ respectively. If $\iota > 0$, the descriptions are incompatible.

Remark 4.5. The minimum over compression schemes is necessary because a single description may admit multiple valid compression schemes (different choices of primitives that yield the same deductive closure). The incompatibility degree is a property of the description pair, not of any particular compression scheme. That $\iota > 0$ after minimisation means that *every* pair of compression schemes for the two descriptions exhibits at least one shared observable with different primitive-derived status.

Proof. Each observable f contributing to α_{12} is primitive in $\mathcal{D}_1$ and derived in $\mathcal{D}_2$. By the argument of Theorem 4.3, each such observable generates at least one contradictory pair of statements. Similarly for each observable contributing to α_{21} . The

contradictions are independent (they concern different observables), so the total is at least $\alpha_{12} + \alpha_{21}$. $\square$

4.4 Physical Illustration: Tunnelling vs. Electrostatics

Example 4.6 (Quantum Tunnelling vs. Classical Electrostatics). Consider a particle of energy E encountering a potential barrier of height $V > E$.

Domain Ω_1 (classical electrostatics): The state space is the classical phase space. The primitive observables include the particle's energy E , the barrier height V , and the particle's position x . The description $\mathcal{D}_1$ asserts: "If $E < V$, then $P(\text{particle beyond barrier}) = 0$." The barrier-crossing probability is a *primitive* consequence of energy conservation.

Domain Ω_2 (quantum mechanics): The state space is a Hilbert space. The primitive observables include the wavefunction ψ and the Hamiltonian $\hat{H}$. The description $\mathcal{D}_2$ *derives* the barrier-crossing probability from the Schrödinger equation: $P(x > x_{\text{barrier}}) = \int_{x_{\text{barrier}}}^{\infty} |\psi(x)|^2 dx > 0$.

The shared observable is the barrier-crossing probability. In $\mathcal{D}_1$, this is determined by energy conservation alone and equals zero. In $\mathcal{D}_2$, it is derived from wavefunction dynamics and is positive. The incompatibility degree is $\iota \geq 1$.

The descriptions are incompatible: $\mathcal{D}_1 \vdash P = 0$ and $\mathcal{D}_2 \vdash P > 0$ for the same physical situation.

Strategic Purpose

The information-theoretic compression argument establishes the incompatibility thesis through the most transparent mechanism: the choice of what counts as fundamental. Any valid, compressive description must treat some observables as primitive and others as derived. When two descriptions of overlapping domains make different choices—as they must, since they compress different regularities—the resulting structural commitments contradict each other on the overlap. This proof method operates in the information-theoretic framework and yields quantitative bounds on the degree of incompatibility.

Self-Critique and Resolution

Critique 1: The “genericity” argument in the proof relies on an implicit **measure-theoretic claim**. The assertion that W' is non-empty “generically” requires formalisation.

Resolution. The genericity claim is made precise in Thesis IV (the topological argument, §7), where we show that the set of contradictory states has positive measure and is topologically residual. The present argument establishes the existence of incompatibility under the stated hypotheses; the topological argument strengthens the quantitative claim.

Critique 2: The proof assumes $P_1 \neq P_2$. What if both descriptions choose identical primitives?

Resolution. If $P_1 = P_2$, then Non-Degeneracy (Axiom 4) requires $R_1 \neq R_2$: the descriptions derive different relationships among the same primitives. This case is treated in Thesis II (§5), where we show that non-isomorphic reduction maps over the same primitives also produce incompatibility through orthogonal structural commitments.

Critique 3: In the tunnelling example, one might argue quantum mechanics simply supersedes classical electrostatics.

Resolution. Classical electrostatics is not wrong within its domain—it makes precise, confirmed predictions for macroscopic electromagnetic phenomena. The incompatibility is between two *valid* descriptions. The reductionist hope that the wider theory subsumes the narrower is examined in Antithesis II (§10), where we show that subsumption requires abandoning the narrower theory’s structural commitments—which are precisely what make it compressive and valid in its domain.

5 Thesis II: The Logical Orthogonality Argument

The compression argument of the preceding section showed that descriptions choosing different primitive observables must conflict. The present argument reaches the incompatibility thesis by a different route: it shows that the *axioms* required to describe different domains are not merely different but *logically orthogonal*—they make structural commitments that are mutually exclusive within any single consistent formal system.

5.1 Orthogonality of Axiom Systems

Definition 5.1 (Axiom Orthogonality). Two axiom sets $\mathcal{A}_1$ and $\mathcal{A}_2$ over a shared language $\mathcal{L}_\cap$ are *orthogonal* if there exists a sentence $\varphi \in \mathcal{L}_\cap$ such that:

$$\mathcal{A}_1 \vdash \varphi \quad \text{and} \quad \mathcal{A}_2 \vdash \neg\varphi.$$

They are *weakly orthogonal* if there exists a sentence φ such that $\mathcal{A}_1 \vdash \varphi$ and $\varphi \notin \text{Th}(\mathcal{A}_2)$ (i.e., $\mathcal{A}_2$ neither proves nor refutes φ).

Orthogonality is a strong condition: the two axiom sets derive contradictory consequences in a shared language. Weak orthogonality is the condition that one axiom set derives a statement the other cannot address. The incompatibility thesis requires orthogonality, not merely weak orthogonality.

Theorem 5.2 (Orthogonality of Scale-Specific Descriptions). *Let $\mathcal{S}$ satisfy Axioms 1–4 with complexity above $C_{\text{threshold}}$. Let $\mathcal{D}_1 = (\mathcal{L}_1, \mathcal{A}_1, \vdash)$ and $\mathcal{D}_2 = (\mathcal{L}_2, \mathcal{A}_2, \vdash)$ be valid descriptions of overlapping domains Ω_1, Ω_2 . Then $\mathcal{A}_1$ and $\mathcal{A}_2$ are orthogonal over the shared language $\mathcal{L}_\cap = \mathcal{L}_1 \cap \mathcal{L}_2$.*

Proof. We give two proofs: a rigorous model-theoretic proof and an illustrative case analysis.

Primary proof (model-theoretic). Consider the restrictions of $\mathcal{D}_1$ and $\mathcal{D}_2$ to the shared language $\mathcal{L}_\cap$ over the overlap region $\Sigma_\cap$. By Non-Degeneracy, $\text{Th}(\mathcal{D}_1|_{\Sigma_\cap}) \neq$

$\text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$: the restrictions have different deductive closures. (If they agreed on the overlap, the descriptions would differ only on observables outside the overlap, but Descriptive Completeness requires both to address $\Sigma_\cap$'s observables, so any non-isomorphism must manifest in the overlap.)

Since the two restricted theories have different deductive closures, there exists a sentence $\varphi \in \mathcal{L}_\cap$ such that $\varphi \in \text{Th}(\mathcal{D}_1|_{\Sigma_\cap})$ and $\varphi \notin \text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$.

We must strengthen this from weak orthogonality to orthogonality. The sentence φ , being in $\mathcal{L}_\cap$, constrains some stable observable $f^* \in \mathcal{O}_\cap$ —for instance, φ might assert a bound on f^* 's range, a functional relationship involving f^* , or a symmetry property of the dynamics restricted to f^* .

By Descriptive Completeness (Axiom 1), $\mathcal{D}_2$ must make a non-trivial prediction about f^* —specifically, there exists some sentence ψ about f^* in $\text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$ with $\psi \neq \top$ (the trivial tautology).

Critical step: We show $\mathcal{D}_2 \vdash \neg\varphi$. Since φ constrains f^* (asserting a specific structural property), and $\mathcal{D}_2$ makes its own non-trivial prediction ψ about f^* , we need that ψ and φ are logically incompatible.

This follows from the Compression Imperative. Both $\mathcal{D}_1$ and $\mathcal{D}_2$ are compressive: they derive specific, constraining structural relationships. The sentence φ is a non-trivial consequence of $\mathcal{D}_1$'s compression scheme (its structural commitments about f^*). Description $\mathcal{D}_2$, using a *different* compression scheme (by Non-Degeneracy), derives *different* structural commitments about f^* . Specifically, if $\mathcal{D}_1$ asserts $f^* = g_1(P_1)$ and $\mathcal{D}_2$ asserts $f^* = g_2(P_2)$ with $g_1 \neq g_2$, then the structural assertions are contradictory: $\mathcal{D}_2 \vdash (f^* = g_2(P_2))$, which combined with $g_1 \neq g_2$ yields $\mathcal{D}_2 \vdash \neg(f^* = g_1(P_1))$, i.e., $\mathcal{D}_2 \vdash \neg\varphi$.

If $\mathcal{D}_2$ does not derive $\neg\varphi$ and does not derive φ , then $\mathcal{D}_2$ is silent about the specific structural relationship φ asserts. But φ is a consequence of a compression scheme applied to f^* , and $\mathcal{D}_2$ has its own compression scheme that addresses f^* . Two different compression schemes for the same observable necessarily entail different functional relationships (by the functional independence condition in the compression scheme definition), and these are logically incompatible. Therefore $\mathcal{D}_2$ derives $\neg\varphi$.

Illustrative case analysis. To make the abstract argument concrete, we note that the structural properties where descriptions typically conflict fall into three natural categories—though the result does not depend on this categorisation:

Case 1: Different functional dependencies. $\mathcal{D}_1$ asserts $f^* = g_1(P_1)$; $\mathcal{D}_2$ asserts $f^* = g_2(P_2)$ with $g_1 \neq g_2$.

Case 2: Different modal status. $\mathcal{D}_1$ asserts f^* is deterministic; $\mathcal{D}_2$ asserts f^* is stochastic.

Case 3: Different symmetry properties. $\mathcal{D}_1$ asserts time-reversibility; $\mathcal{D}_2$ asserts irreversibility.

These illustrate the theorem but the proof rests on the model-theoretic argument, not on the exhaustiveness of this list. $\square$

5.2 Alternative Proof via Model Theory

We provide a second proof using model-theoretic methods, which avoids the case analysis.

Alternative proof of Theorem 5.2. Consider the class of models of each description restricted to the overlap. Let $\text{Mod}_\cap(\mathcal{D}_1) = \{\mathfrak{M} \models \mathcal{D}_1|_{\Sigma_\cap}\}$ and $\text{Mod}_\cap(\mathcal{D}_2) = \{\mathfrak{M} \models \mathcal{D}_2|_{\Sigma_\cap}\}$ be the classes of structures on $\Sigma_\cap$ satisfying each description.

By Non-Degeneracy, $\text{Th}(\mathcal{D}_1|_{\Sigma_\cap}) \neq \text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$: the restrictions of the two theories to the overlap have different deductive closures. (If they agreed on the overlap, the descriptions would be compatible on $\Sigma_\cap$, and the only non-isomorphism would concern observables outside the overlap—but Descriptive Completeness requires both to address $\Sigma_\cap$ ’s observables.)

Since the two restricted theories have different deductive closures, there exists a sentence φ in the shared language such that $\varphi \in \text{Th}(\mathcal{D}_1|_{\Sigma_\cap})$ and $\varphi \notin \text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$.

We must strengthen this from weak orthogonality to orthogonality. Suppose $\neg\varphi \notin \text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$ either—that is, $\mathcal{D}_2$ is silent about φ . But φ concerns a shared observable (it is in $\mathcal{L}_\cap$), and Descriptive Completeness requires $\mathcal{D}_2$ to make non-trivial predictions about all shared observables. If φ is a sentence about a stable observable f^* (e.g., a constraint on f^* ’s range, a functional dependency, or a symmetry property), then $\mathcal{D}_2$ ’s silence about φ violates Completeness.

Therefore, $\mathcal{D}_2$ must either prove φ or prove $\neg\varphi$. Since $\varphi \notin \text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$, we have $\neg\varphi \in \text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$. Hence $\mathcal{A}_1 \vdash \varphi$ and $\mathcal{A}_2 \vdash \neg\varphi$: orthogonality. $\square$

5.3 Physical Illustration: Reversibility vs. Irreversibility

Example 5.3 (Microscopic Reversibility vs. Thermodynamic Irreversibility). Consider a gas of N particles in a box.

Domain Ω_1 (Hamiltonian mechanics): The state space is the $6N$ -dimensional phase space. The dynamics is governed by Hamilton’s equations, which are time-reversible: for every trajectory $\sigma(t)$, the time-reversed trajectory $\sigma(-t)$ (with reversed momenta) is also a valid trajectory. The description $\mathcal{D}_1$ derives $\varphi \equiv \ulcorner \text{the dynamics is time-reversible} \urcorner$.

Domain Ω_2 (thermodynamics): The state space is the space of macroscopic variables (temperature, pressure, volume, entropy). The dynamics is governed by the laws of thermodynamics, including the second law: entropy does not decrease in an isolated system. The description $\mathcal{D}_2$ derives $\neg\varphi \equiv \ulcorner \text{the dynamics is irreversible} \urcorner$.

The overlap region is the same physical system—the gas—viewed at two different scales. The shared observable includes the macroscopic state of the gas. The descriptions are orthogonal: $\mathcal{D}_1 \vdash \varphi$ and $\mathcal{D}_2 \vdash \neg\varphi$.

This is not a matter of one theory being “more correct.” Hamiltonian mechanics is precisely validated at the microscopic level; thermodynamics is precisely validated at the macroscopic level. The irreversibility of thermodynamics is not an approximation of reversible dynamics—it is a genuine structural property of the macroscopic domain that *contradicts* the structural property of the microscopic domain.

Strategic Purpose

The logical orthogonality argument reaches incompatibility from a different angle than the compression argument. Where the compression argument showed that different *choices of primitives* conflict, the orthogonality argument shows that different *axiom systems* derive contradictory consequences about shared observables. The model-theoretic proof avoids case analysis and grounds the result in the structural fact that non-isomorphic theories on a shared domain must disagree on some sentence—and Descriptive Completeness prevents either theory from remaining silent. The two proof methods (case analysis and model theory) reinforce each other.

Self-Critique and Resolution

Critique 1: The case analysis in the first proof may not be exhaustive. Are functional dependencies, modal status, and symmetry properties the only ways descriptions can differ?

Resolution. The model-theoretic proof avoids this concern entirely by showing that *any* difference in deductive closure suffices. The case analysis is provided for concreteness and physical motivation, not as the logical backbone. The result rests on the model-theoretic argument.

Critique 2: The role of Descriptive Completeness in strengthening weak orthogonality to full orthogonality is subtle.

Resolution. This is the key step, and it deserves emphasis. Without Descriptive Completeness, two descriptions could differ on a sentence φ while one remains agnostic—yielding weak orthogonality but not contradiction. Descriptive Completeness forbids this agnosticism: if φ concerns a shared stable observable, the description *must* take a position. This is the axiom’s essential function in the proof, and it explains why the axiom was needed.

Critique 3: The reversibility example is standard and much-discussed. Is it adding anything new?

Resolution. The example illustrates the theorem; it is not evidence for it. Its value is familiarity: every physicist recognises the tension between microscopic reversibility and macroscopic irreversibility. The theorem’s contribution is to show that this tension is not a puzzle to be resolved but a *necessary structural feature* of any sufficiently complex system.

6 Thesis III: The Category-Theoretic Non-Existence-of-Limit Argument

The preceding arguments exploited the content of descriptions (what they compress, what they assert). The present argument operates at the structural level: it shows that the *category of descriptions* of a sufficiently complex system lacks a limit object—

a universal description from which all others can be recovered. This is the formal expression of the claim that no unified theory exists.

6.1 The Category of Descriptions

Definition 6.1 (Description Category). Given a system $\mathcal{S}$ satisfying the four axioms, define the category $\mathbf{Desc}(\mathcal{S})$ as follows:

- *Objects*: valid descriptions $\mathcal{D}_i$ of domains $\Omega_i \subseteq \mathcal{S}$;
- *Morphisms*: a morphism $\mu : \mathcal{D}_i \rightarrow \mathcal{D}_j$ exists if and only if $\Omega_i \subseteq \Omega_j$ (the domain of $\mathcal{D}_i$ is contained in the domain of $\mathcal{D}_j$) and there is a theory interpretation $\mu^* : \text{Th}(\mathcal{D}_i) \rightarrow \text{Th}(\mathcal{D}_j)$ that preserves truth on Ω_i : for every $\varphi \in \text{Th}(\mathcal{D}_i)$, the image $\mu^*(\varphi)$ holds in Ω_i whenever φ does.

Morphisms point from smaller domains to larger ones: a morphism $\mu : \mathcal{D}_i \rightarrow \mathcal{D}_j$ says “ $\mathcal{D}_j$ can interpret the predictions of $\mathcal{D}_i$ within its own framework.” This is the standard direction for a colimit (direct limit) construction. Composition of morphisms is composition of theory interpretations; identity morphisms are identity interpretations.

A *unified description* would be a colimit of this diagram: a description $\mathcal{D}_\infty$ with morphisms from each $\mathcal{D}_i$ to $\mathcal{D}_\infty$, such that $\mathcal{D}_\infty$ is the “smallest” consistent description containing all local descriptions.

Definition 6.2 (Colimit of the Description Category). A *colimit* of the diagram $\mathbf{Desc}(\mathcal{S})$ is a description $\mathcal{D}_\infty$ together with morphisms $\iota_i : \mathcal{D}_i \rightarrow \mathcal{D}_\infty$ for every object $\mathcal{D}_i$, such that for every morphism $\mu : \mathcal{D}_i \rightarrow \mathcal{D}_j$, the diagram commutes: $\iota_j \circ \mu = \iota_i$. Furthermore, $\mathcal{D}_\infty$ is universal: any other description with the same property factors uniquely through $\mathcal{D}_\infty$.

The existence of such a colimit would mean: there is a single consistent theory into which all valid local descriptions can be faithfully embedded, with the embeddings mutually consistent on overlaps.

Theorem 6.3 (Non-Existence of Colimit). *Let $\mathcal{S}$ satisfy Axioms 1–4 with complexity above $C_{\text{threshold}}$, and let $\mathcal{D}_1, \mathcal{D}_2$ be orthogonal descriptions (as established by Theorem 5.2) of overlapping domains. Then $\mathbf{Desc}(\mathcal{S})$ has no colimit.*

Proof. Suppose, for contradiction, that a colimit $\mathcal{D}_\infty = (\mathcal{L}_\infty, \mathcal{A}_\infty, \vdash_\infty)$ exists with embedding morphisms $\iota_1 : \mathcal{D}_1 \rightarrow \mathcal{D}_\infty$ and $\iota_2 : \mathcal{D}_2 \rightarrow \mathcal{D}_\infty$.

By the definition of morphism, ι_1 embeds $\text{Th}(\mathcal{D}_1)$ into $\text{Th}(\mathcal{D}_\infty)$: for every $\varphi \in \text{Th}(\mathcal{D}_1)$, $\iota_1^*(\varphi) \in \text{Th}(\mathcal{D}_\infty)$. Similarly for ι_2 .

By Theorem 5.2, there exists $\varphi \in \mathcal{L}_\cap$ such that $\mathcal{D}_1 \vdash \varphi$ and $\mathcal{D}_2 \vdash \neg\varphi$.

The commutativity condition requires that on the overlap region, the embeddings agree: for any sentence ψ about the overlap, $\iota_1^*(\psi)$ and $\iota_2^*(\psi)$ must have the same truth value in $\mathcal{D}_\infty$. This is because both $\mathcal{D}_1$ and $\mathcal{D}_2$ are connected to the overlap via restriction morphisms, and commutativity requires these paths to agree.

Applying this to φ : $\iota_1^*(\varphi) \in \text{Th}(\mathcal{D}_\infty)$ (since $\varphi \in \text{Th}(\mathcal{D}_1)$), and $\iota_2^*(\neg\varphi) \in \text{Th}(\mathcal{D}_\infty)$ (since $\neg\varphi \in \text{Th}(\mathcal{D}_2)$). By commutativity on the overlap, $\iota_2^*(\neg\varphi) = \neg\iota_1^*(\varphi)$, so $\mathcal{D}_\infty$ derives both $\iota_1^*(\varphi)$ and $\neg\iota_1^*(\varphi)$, contradicting the consistency of $\mathcal{D}_\infty$.

The only escape is to drop the commutativity requirement—but without commutativity, the “colimit” does not consistently unify the local descriptions; it merely contains disconnected copies of them. Such a structure is a coproduct (disjoint union), not a colimit, and does not constitute unification in any meaningful sense. $\square$

Key Insight

The category-theoretic argument formalises the intuition that “no master theory contains all local theories as special cases.” In the language of category theory, this is the statement that the diagram of valid descriptions has no limit. The obstruction is not computational (we could not find the limit) but structural (the limit would be inconsistent). This is a stronger impossibility than practical infeasibility.

6.2 Physical Illustration: The Absence of a “Theory of Everything”

The non-existence of the limit is a formal version of the observation that no single theoretical framework in physics successfully encompasses all known phenomena while preserving the structural commitments of each sub-theory. General relativity and quantum field theory are each valid within their domains; their “unification” (quantum gravity) remains elusive not because of insufficient ingenuity but because the structural commitments of the two frameworks are orthogonal: GR requires smooth spacetime geometry; QFT requires a fixed background with quantum fluctuations. The theorem predicts that this is not a contingent difficulty but a necessary feature of any system complex enough to require both descriptions.

Strategic Purpose

The categorical argument reaches incompatibility at the structural level, without examining the specific content of any description. It shows that the *architecture* of a sufficiently complex system—the web of local descriptions and the morphisms between them—is structurally incompatible with the existence of a global description. This is the most abstract of the five arguments and the one most directly relevant to the “Theory of Everything” programme.

Self-Critique and Resolution

Critique 1: The category $\text{Desc}(\mathcal{S})$ is defined ad hoc for this paper. Is it a well-known categorical construction?

Resolution. The construction is modelled on the category of theories and interpretations, which is standard in mathematical logic [Mac Lane, 1971]. The specific

adaptation to physical descriptions is new. The categorical machinery is used as a language for stating a precise result, not as a source of hidden assumptions.

Critique 2: The “faithful interpretation” requirement may be too strong.

Resolution. If we drop faithfulness, the limit may exist—but it would be a theory that equivocates on the meaning of shared observables. Such a “unified” theory would assign different meanings to the same physical quantity depending on which local description is being recovered. This is linguistically possible but physically vacuous: it amounts to renaming the contradiction rather than resolving it. The point is examined in Antithesis III (§11).

7 Thesis IV: The Topological Obstruction Argument

The preceding arguments established that incompatibility exists. The present argument shows that it is *generic*—that the set of states where descriptions conflict has positive measure and is topologically unavoidable. This elevates incompatibility from a logical possibility to a structural feature of the description space.

7.1 The Description Manifold

We equip the space of descriptions with topological structure. To ensure rigour, we restrict attention to *parametric families* of descriptions—descriptions that share an axiom structure and differ only in the values of finitely many real parameters (coupling constants, cutoff scales, coefficients). This covers the physically relevant case of effective theories at different scales and avoids the difficulty of defining a topology on the space of all possible axiom systems.

Definition 7.1 (Parametric Description Space). Given an overlap region $\Sigma_\cap$ with shared observables $\mathcal{O}_\cap = \{f_1, \dots, f_n\}$, and a fixed axiom template $\mathcal{A}(\theta)$ parametrised by $\theta \in \Theta \subseteq \mathbb{R}^k$, the *parametric description space* is $\mathcal{M} = \Theta$ with the standard Euclidean topology. Each $\theta \in \Theta$ yields a description $\mathcal{D}_\theta$ with axioms $\mathcal{A}(\theta)$.

For descriptions differing in axiom *structure* (not just parameters), we use the discrete topology, where every set is open. The density argument in this case operates through structural perturbations (changing which observables are primitive) rather than through continuous parameter variation.

Definition 7.2 (Compatibility Region). The *compatibility region* $\mathcal{C} \subseteq \mathcal{M} \times \mathcal{M}$ is the set of pairs of descriptions $(\mathcal{D}_{\theta_1}, \mathcal{D}_{\theta_2})$ that are compatible in the sense of Definition 2.7. The *incompatibility region* is $\mathcal{I} = (\mathcal{M} \times \mathcal{M}) \setminus \mathcal{C}$.

Theorem 7.3 (Topological Genericity of Incompatibility). *Let $\mathcal{S}$ satisfy the four axioms with complexity above $C_{\text{threshold}}$. In the parametric description space:*

1. *Within any parametric family $\mathcal{M} = \Theta$ containing descriptions of two overlapping domains with non-isomorphic axiom structures, the incompatibility region $\mathcal{I}$ is open in $\Theta \times \Theta$.*

2. The compatibility region $\mathcal{C}$ is closed and, for generic parametric families satisfying the four axioms, has empty interior.
3. For descriptions differing in axiom structure (discrete topology), every pair of non-isomorphic descriptions of overlapping domains is incompatible—the result is absolute, not merely generic.

Proof. Part (1): Let $(\mathcal{D}_{\theta_1}, \mathcal{D}_{\theta_2}) \in \mathcal{I}$ be an incompatible pair. By Theorem 5.2, there exists φ such that $\mathcal{D}_{\theta_1} \vdash \varphi$ and $\mathcal{D}_{\theta_2} \vdash \neg\varphi$. Derivability from a finite axiom set $\mathcal{A}(\theta)$ is a property that depends on θ through the truth values of finitely many conditions. For each sentence φ , the set $\{\theta : \mathcal{A}(\theta) \vdash \varphi\}$ is defined by finitely many polynomial inequalities (in the case of linear or polynomial axiom templates) and is therefore open or a union of open and closed sets. In either case, the set $\{(\theta_1, \theta_2) : \mathcal{A}(\theta_1) \vdash \varphi \text{ and } \mathcal{A}(\theta_2) \vdash \neg\varphi\}$ is open. Since $\mathcal{I}$ is the union of such sets over all possible φ , it is open.

Part (2): $\mathcal{C}$ is the complement of an open set, hence closed. For a parametric family that includes descriptions with genuinely different compression schemes (as required by Non-Degeneracy), compatible pairs require the parameter values to satisfy specific algebraic conditions that make the two compression schemes yield identical structural commitments on the overlap. These conditions generically define a set of positive codimension (i.e., measure zero) in $\Theta \times \Theta$.

Part (3): For descriptions with different axiom structures, the incompatibility follows directly from Theorem 5.2 without any topological argument—every such pair is incompatible. $\square$

7.2 Cohomological Interpretation

The topological result admits a cohomological reading. The descriptions $\mathcal{D}_i$ can be viewed as *local sections* of a “theory sheaf” over the space of domains. The incompatibility of descriptions on overlaps corresponds to a non-trivial cohomology class: the local sections cannot be glued into a global section.

Remark 7.4 (Cohomology of Descriptions). Define a presheaf $\mathcal{F}$ on the poset of domains (ordered by inclusion) by assigning to each domain Ω_i the set of valid descriptions $\mathcal{F}(\Omega_i) = \{\mathcal{D} : \mathcal{D} \models \Omega_i\}$, with restriction maps given by the theory-restriction operation. The incompatibility of descriptions on overlaps means that this presheaf fails the gluing axiom: local sections (valid descriptions on individual domains) cannot be assembled into a global section (a valid description on the entire system). In sheaf-theoretic language, $\mathcal{F}$ is a presheaf that is not a sheaf, and the obstruction to being a sheaf is measured by the first Čech cohomology group $\check{H}^1(\{\Omega_i\}, \mathcal{F})$. When this group is non-trivial, no global description exists.

This cohomological perspective connects the incompatibility thesis to well-known obstructions in mathematics: the inability to define a global coordinate system on a manifold (the Euler class), the impossibility of a global section of a non-trivial bundle, and the contextuality obstructions in quantum mechanics.

Strategic Purpose

The topological argument establishes that incompatibility is not a marginal or exceptional phenomenon but the *generic* condition in the description space. Compatible descriptions form a measure-zero, nowhere-dense set—they are the exception, not the rule. The cohomological interpretation connects this result to deep structural obstructions in mathematics, showing that the incompatibility of descriptions is analogous to the impossibility of globalising local structures in topology and geometry.

Self-Critique and Resolution

Critique 1: The restriction to parametric families limits the generality of the topological result.

Resolution. This is acknowledged. The topological argument applies rigorously within parametric families, which cover the physically relevant case (effective theories parametrised by coupling constants and cutoffs). For descriptions with different axiom structures, the incompatibility is absolute (Part 3 of the theorem), not merely generic—making the topological argument unnecessary in that case. The restriction ensures that the topology is well-defined and the proof is rigorous, at the cost of not covering all conceivable description spaces.

Critique 2: The cohomological interpretation is suggestive but not fully developed.

Resolution. A complete sheaf-theoretic treatment would require specifying the site, the restriction maps, and computing the cohomology explicitly. This is beyond the scope of the present paper but is noted as a direction for future work. The remark is included to connect the result to existing mathematical language, not to claim a new sheaf-theoretic result.

8 Thesis V: The Predictive Trade-Off Argument

The four preceding arguments have been structural: they concerned what descriptions *assert*, not how well they *perform*. The present argument addresses the pragmatic dimension directly. It shows that any unified description covering both domains must suffer predictive degradation in at least one—a quantifiable loss of accuracy that renders the unified description inferior to the local descriptions it replaces.

8.1 Predictive Accuracy and the Unification Penalty

Definition 8.1 (Predictive Accuracy). Given a description $\mathcal{D}$ valid for domain Ω and a stable observable $f \in \mathcal{O}_\Omega$, the *predictive accuracy* of $\mathcal{D}$ for f is

$$\mathcal{P}(\mathcal{D}, f) = 1 - \frac{1}{|\Sigma_\Omega|} \sum_{\sigma \in \Sigma_\Omega} |f(\sigma) - \hat{f}_{\mathcal{D}}(\sigma)|^2,$$

where $\hat{f}_{\mathcal{D}}(\sigma)$ is the value predicted by $\mathcal{D}$ for observable f at state σ . For a valid description (Definition 2.6), $\hat{f}_{\mathcal{D}}(\sigma) = f(\sigma)$ wherever $\mathcal{D}$ makes a prediction, so $\mathcal{P}(\mathcal{D}, f) = 1$ for all observables within the domain.

Definition 8.2 (Unification Attempt). A *unification attempt* for descriptions $\mathcal{D}_1, \mathcal{D}_2$ of overlapping domains Ω_1, Ω_2 is a description $\mathcal{D}_U$ defined on $\Omega_1 \cup \Omega_2$ that:

1. is internally consistent;
2. satisfies the Compression Imperative for $\Omega_1 \cup \Omega_2$;
3. makes predictions for all stable observables in $\mathcal{O}_1 \cup \mathcal{O}_2$.

Note that a unification attempt is required to be consistent and compressive but is *not* required to be valid—we are asking whether a valid unification exists, and this theorem shows it cannot.

Theorem 8.3 (Predictive Trade-Off). *Let $\mathcal{D}_1, \mathcal{D}_2$ be valid, compressive descriptions of overlapping domains Ω_1, Ω_2 in a system satisfying the four axioms with complexity above $C_{\text{threshold}}$. For any consistent unification attempt $\mathcal{D}_U$:*

$$\min(\mathcal{P}(\mathcal{D}_U, f)|_{\Omega_1}, \mathcal{P}(\mathcal{D}_U, f)|_{\Omega_2}) < 1$$

for at least one shared observable $f \in \mathcal{O}_{\cap}$. That is, the unified description must sacrifice perfect accuracy in at least one domain.

Proof. By Theorem 5.2, there exists a sentence φ about the overlap such that $\mathcal{D}_1 \vdash \varphi$ and $\mathcal{D}_2 \vdash \neg\varphi$. Both φ and $\neg\varphi$ correspond to empirically confirmed predictions within their respective domains.

$\mathcal{D}_U$ must be consistent, so it cannot derive both φ and $\neg\varphi$. It must choose: either $\mathcal{D}_U \vdash \varphi$, or $\mathcal{D}_U \vdash \neg\varphi$, or $\mathcal{D}_U$ remains silent on φ .

Case 1: $\mathcal{D}_U \vdash \varphi$. Then $\mathcal{D}_U$ disagrees with $\mathcal{D}_2$ on φ . Since $\mathcal{D}_2 \vdash \neg\varphi$ was empirically confirmed in Ω_2 , the unified description makes a false prediction in Ω_2 . Its predictive accuracy for the observable associated with φ drops below 1 in Ω_2 .

Case 2: $\mathcal{D}_U \vdash \neg\varphi$. By symmetric reasoning, the unified description's accuracy drops below 1 in Ω_1 .

Case 3: $\mathcal{D}_U$ is silent on φ . By Descriptive Completeness (Axiom 1), a valid description of $\Omega_1 \cup \Omega_2$ must address all stable observables. If $\mathcal{D}_U$ is silent on φ , it violates Completeness and is therefore not a valid description of the combined domain. Alternatively, if we relax Completeness and allow the silence, the unified description loses predictive power: it makes fewer predictions than the local descriptions it replaces.

In all cases, the unified description either loses accuracy or loses coverage—it is either wrong somewhere or silent somewhere. Perfect accuracy in both domains is impossible. $\square$

8.2 The Kolmogorov Complexity Bound

The predictive trade-off can be restated in information-theoretic language.

Corollary 8.4 (Complexity Cost of Unification). *For any consistent unification attempt $\mathcal{D}_U$ of orthogonal descriptions $\mathcal{D}_1, \mathcal{D}_2$:*

$$K(\mathcal{D}_U) \geq K(\mathcal{D}_1) + K(\mathcal{D}_2) + K(\text{conflict resolution}),$$

where $K(\text{conflict resolution}) > 0$ is the Kolmogorov complexity of specifying, for each contradictory pair $(\varphi, \neg\varphi)$, which version the unified description adopts and how the resulting errors are distributed.

This means that the unified description is necessarily *more complex* (in the information-theoretic sense) than either local description, and the excess complexity does not purchase additional predictive power—it is spent entirely on managing the contradictions. The unified description is computationally more expensive and predictively no better (and typically worse) than the pair of local descriptions.

Strategic Purpose

The predictive trade-off argument addresses the most pragmatic objection to the incompatibility thesis: “even if descriptions are formally incompatible, perhaps a unified theory can reproduce their predictions in practice.” The answer is no: the unified theory must either make false predictions in at least one domain or remain silent where the local theories speak. Combined with the Kolmogorov complexity bound, this means unification is not merely logically impossible but *practically counterproductive*—it costs more and delivers less.

Self-Critique and Resolution

Critique 1: The predictive accuracy metric assumes a finite state space. For continuous state spaces, the sum should be an integral.

Resolution. The metric generalises straightforwardly to continuous state spaces via integration against a probability measure on Σ_Ω . The theorem’s logic does not depend on the metric’s precise form: the key point is that the unified description must make at least one false prediction, regardless of how accuracy is measured.

Critique 2: The Kolmogorov complexity bound is stated informally.

Resolution. A fully rigorous statement would require fixing a universal Turing machine and carefully defining the “conflict resolution” programme. The informal statement captures the essential point: unification has an irreducible information-theoretic cost. Making this fully rigorous within algorithmic information theory is noted as future work.

Critique 3: A unified description might adopt a “contextual” approach—applying φ in Ω_1 and $\neg\varphi$ in Ω_2 .

Resolution. This is precisely what physicists do in practice (“use quantum mechanics here, classical mechanics there”). But a contextual theory is not a unified

description in the sense of Definition 2.7: it is a meta-theory that selects among local descriptions based on context. It does not derive both φ and $\neg\varphi$ from a single consistent axiom set; it switches between axiom sets. The Compression Imperative requires the unified description to be shorter than a concatenation of the local descriptions, and the contextual approach—which must additionally encode the switching rules—fails this requirement. This point is developed in Antithesis II (§10).

9 Antithesis I: The Trivial-Domain Construction

We now turn the investigation on its head. Having presented five arguments for the incompatibility thesis, we subject it to sustained critique by genuinely attempting to prove its negation.

9.1 The Claim

Proposition 9.1 (Full Compatibility in Simple Systems). *There exist systems satisfying three of the four axioms in which all valid descriptions are mutually compatible.*

Construction. Consider a system $\mathcal{S}_{\text{triv}}$ with a single domain Ω (so that $\delta(\mathcal{S}) = 1$). Since Non-Degeneracy (Axiom 4) requires $\delta \geq 2$, this system violates Non-Degeneracy. All valid descriptions of Ω are equivalent up to isomorphism, so they are trivially compatible.

Alternatively, consider a system with two domains Ω_1, Ω_2 that do not overlap ($\Omega_1 \cap \Omega_2 = \emptyset$), violating Scale Coupling (Axiom 3). The descriptions of the two domains are about disjoint sets of observables and cannot conflict. $\square$

9.2 Does This Refute the Thesis?

No. The constructions succeed only by violating the axioms that define the regime of interest. A system with a single domain is not complex enough to exhibit the phenomenon. A system with non-overlapping domains has no arena for contradiction.

Theorem 9.2 (Minimum Conditions for Incompatibility). *If a system satisfies all four axioms (Completeness, Compression, Coupling, Non-Degeneracy) with complexity above $C_{\text{threshold}}$, then incompatible descriptions necessarily exist.*

Proof. This is the conjunction of Theorems 4.3, 5.2, 6.3, 7.3, and 8.3, each established independently. $\square$

The antithesis identifies a sharp boundary: compatibility is possible only when the system lacks the structural features (overlapping domains, genuine diversity, compressive descriptions) that define non-trivial complexity.

Strategic Purpose

The trivial-domain antithesis plays the same role as the trivial-system construction in Article 1: it succeeds in degenerate cases and thereby sharpens the thesis by identifying exactly which conditions are necessary.

Self-Critique and Resolution

Critique 1: The “sharp boundary” claim depends on the axioms being the right ones.

Resolution. The axioms are stated as hypotheses, not as metaphysical truths. A reader who rejects any axiom is invited to identify which physical systems satisfy only the remaining three and whether those systems exhibit the complexity threshold phenomenon.

10 Antithesis II: The Hierarchy-of-Approximations Argument

The second antithesis is the most physically motivated: the reductionist claim that one theory (the “more fundamental” one) subsumes the other as an approximation.

10.1 The Claim

Proposition 10.1 (Compatibility via Limiting Relations). *If $\mathcal{D}_2$ reduces to $\mathcal{D}_1$ in a suitable limit (e.g., $\hbar \rightarrow 0$, $N \rightarrow \infty$, $c \rightarrow \infty$), the descriptions are compatible: $\mathcal{D}_1$ is a special case of $\mathcal{D}_2$.*

This is the standard reductionist programme. Quantum mechanics “contains” classical mechanics in the limit $\hbar \rightarrow 0$. Statistical mechanics “contains” thermodynamics in the limit $N \rightarrow \infty$.

10.2 Analysis: Does This Succeed?

Theorem 10.2 (Limiting Relations Do Not Eliminate Incompatibility). *The existence of a limiting relation $\mathcal{D}_2 \xrightarrow{\lambda \rightarrow 0} \mathcal{D}_1$ does not establish compatibility in the sense of Definition 2.7.*

Proof. The limiting relation asserts: as a parameter λ approaches a limit value, the predictions of $\mathcal{D}_2$ approach those of $\mathcal{D}_1$. This is an *asymptotic* statement, not a logical one.

The descriptions remain orthogonal at any *finite* value of λ . Quantum mechanics with $\hbar > 0$ (the physical value) predicts tunnelling; classical mechanics predicts no tunnelling. The limit $\hbar \rightarrow 0$ is a mathematical operation that does not correspond to any physical situation— $\hbar$ is a fixed constant of nature.

More formally: the limit $\mathcal{D}_2 \xrightarrow{\lambda \rightarrow 0} \mathcal{D}_1$ is a statement about a family of theories parametrised by λ . At $\lambda = 0$, we recover $\mathcal{D}_1$; at the physical value $\lambda = \lambda_0 > 0$, we have $\mathcal{D}_2$. The two theories $\mathcal{D}_1$ (at $\lambda = 0$) and $\mathcal{D}_2$ (at $\lambda = \lambda_0$) are distinct members of the family. Their compatibility would require a single theory valid at *both* $\lambda = 0$ and $\lambda = \lambda_0$ —but the orthogonal sentence φ (e.g., “tunnelling occurs”) has different truth values at these two parameter values. A consistent theory cannot assign both truth values simultaneously.

The limiting relation provides a *correspondence*—a controlled comparison of predictions—but not a *unification*—a single consistent framework. Correspondence is weaker than compatibility. $\square$

Key Insight

The hierarchy-of-approximations antithesis reveals a crucial distinction: *correspondence is not compatibility*. That one theory approaches another in a limit does not mean they can coexist in a single framework. The limit is a mathematical bridge that connects the theories without merging them. The theories remain logically distinct at every physical parameter value.

Self-Critique and Resolution

Critique 1: Effective field theory *does* provide a systematic framework for theories at different scales.

Resolution. EFT provides a *family* of theories parametrised by scale, with controlled relationships between them. It does not provide a single theory valid at all scales simultaneously. The RG flow that connects the theories is precisely the mechanism that transforms one set of relevant operators into another—changing the effective axioms at each scale. EFT is a formalisation of managed incompatibility, not a resolution of it. This is the insight of the Law of Scale-Specific Principles [Kriger, 2025b].

Critique 2: In practice, physicists successfully use quantum mechanics to predict classical behaviour.

Resolution. They do so by applying decoherence, coarse-graining, and semiclassical approximations—operations that are external to the quantum formalism and require additional structure (environment, observer, scale separation). These operations are precisely the “switching rules” between descriptions. They work excellently in practice, but they do not constitute a unified description; they constitute a meta-framework for navigating between incompatible descriptions.

11 Antithesis III: The Shared-Ontology Covering Argument

The third antithesis asks whether the incompatibility is genuine or merely apparent—whether a deeper ontological level could reconcile the contradictory descriptions.

11.1 The Claim

Proposition 11.1 (Compatibility via Deeper Ontology). *Perhaps both $\mathcal{D}_1$ and $\mathcal{D}_2$ are incomplete projections of a single underlying reality $\mathcal{R}$, and their apparent contradictions arise from the projection, not from $\mathcal{R}$ itself.*

This is the “hidden variables” or “deeper theory” strategy: the descriptions disagree only because each captures a partial aspect of a richer structure.

11.2 Analysis

Theorem 11.2 (The Deeper-Ontology Antithesis Fails or Recurses). *If a deeper ontological level $\mathcal{R}$ exists from which both $\mathcal{D}_1$ and $\mathcal{D}_2$ can be derived as projections, then either:*

1. *$\mathcal{R}$ is itself a valid, compressive description of the combined domain, in which case it inherits the incompatibility problem (it must make a definite prediction about φ and thereby disagree with one of the local descriptions); or*
2. *$\mathcal{R}$ is not compressive (it is at least as complex as the raw data), in which case it violates the Compression Imperative and is not a genuine theory; or*
3. *$\mathcal{R}$ introduces a new domain Ω_3 that overlaps with both Ω_1 and Ω_2 , and the incompatibility thesis applies to $\mathcal{R}$ together with the local descriptions—generating a new level of incompatibility.*

Proof. Case (1): If $\mathcal{R}$ is a valid, compressive description of $\Omega_1 \cup \Omega_2$, then by Theorem 8.3, it cannot perfectly reproduce the predictions of both $\mathcal{D}_1$ and $\mathcal{D}_2$. It must sacrifice accuracy in at least one domain. Therefore $\mathcal{R}$ does not fully “contain” both descriptions—it replaces one of them with a degraded version.

Case (2): If $\mathcal{R}$ ’s specification is at least as long as a brute-force enumeration of all observable values in $\Omega_1 \cup \Omega_2$, it fails the Compression Imperative. It is a database, not a theory.

Case (3): If $\mathcal{R}$ introduces genuinely new structure (new observables, new degrees of freedom), it constitutes a new domain that overlaps with the existing ones. The four axioms may apply to the expanded system $\{\Omega_1, \Omega_2, \Omega_3\}$, and the incompatibility thesis generates new contradictions between $\mathcal{R}$ and the local descriptions. The problem recurses rather than resolves. $\square$

Strategic Purpose

The deeper-ontology antithesis addresses the most philosophically sophisticated objection: “maybe we just haven’t found the right framework yet.” The response is that any “right framework” must itself be either compressive (and therefore subject to the trade-off theorem), non-compressive (and therefore not a theory), or a new level that regenerates the problem. The regress does not terminate.

Self-Critique and Resolution

Critique 1: This argument seems to prove too much—it would rule out *any* theoretical progress.

Resolution. The argument does not rule out progress; it rules out *total unification*. Science progresses by constructing new descriptions that are valid in wider domains—but each new description inherits the incompatibility with descriptions of domains it does not cover. Maxwell’s equations unified electricity and magnetism but are incompatible with quantum electrodynamics at small scales. QED extends the domain but is incompatible with gravity. Each advance is genuine progress; none is total unification.

12 Antithesis IV: Systematic Axiom Relaxation

The final antithesis targets the axioms directly. We ask: if we drop each axiom in turn, does the incompatibility thesis survive?

12.1 Systematic Relaxation

12.1.1 Relaxing Descriptive Completeness

If descriptions are allowed to remain silent about shared observables, they can avoid contradiction by dividing the observables between them. Each description addresses only “its own” observables.

Result: Incompatibility is eliminated—but at the cost of *explanatory gaps*. Neither description covers the full domain; the overlap region contains observables that no description addresses. The system has “dark spots” that no theory illuminates.

12.1.2 Relaxing the Compression Imperative

If “descriptions” are allowed to be as long as the raw data, a trivial “unified theory” can be constructed by concatenating all predictions.

Result: Incompatibility is eliminated—but the “unified theory” is a database, not a theory. It does not compress, does not explain, does not predict beyond what it explicitly lists.

12.1.3 Relaxing Scale Coupling

If domains do not overlap, descriptions have nothing to contradict about.

Result: Incompatibility is eliminated trivially, because the arena for contradiction (the overlap region) does not exist. This is physically unrealistic: the domains of physics are coupled.

12.1.4 Relaxing Non-Degeneracy

If all domains admit the same description, there is no diversity and no contradiction.

Result: Incompatibility is eliminated—but the system is not complex. A single description covering all domains means the system has $\delta = 1$: it is below the complexity threshold.

12.2 Summary of Relaxation Results

Axiom relaxed	Incompatibility removed?	Cost
Descriptive Completeness	Yes	Explanatory gaps
Compression Imperative	Yes	Theory becomes database
Scale Coupling	Yes (trivially)	Domains are isolated
Non-Degeneracy	Yes	System is not complex

Table 1: Effect of relaxing each axiom on the incompatibility thesis.

Key Insight

Every axiom relaxation that eliminates incompatibility does so by sacrificing a feature that is essential to the notion of “describing a complex system.” Incompatibility is the price of having compressive, complete, diverse descriptions of coupled domains. It can be avoided only by giving up one of these features—each of which is a defining characteristic of theoretical science.

Self-Critique and Resolution

Critique 1: The relaxation analysis is symmetric—every axiom is equally necessary. Does this mean the result is trivially dependent on the definitions?

Resolution. The axioms are not arbitrary. Each captures a minimal condition that any reasonable notion of “scientific theory” satisfies: theories must be compressive (Compression), must address their subject matter (Completeness), must engage with shared phenomena (Coupling), and must be genuinely different when describing genuinely different domains (Non-Degeneracy). The result says: if your theoretical practice satisfies these four conditions, incompatibility is unavoidable. Whether to accept these conditions is the reader’s choice.

13 The Independence Investigation

We have attempted to prove incompatibility (five arguments) and to disprove it (four attempts). The thesis prevailed in all non-degenerate cases. We now investigate two further questions: whether an internal observer can determine *which* description governs the overlap region, and whether the compatibility statement is a consequence of the axioms alone or requires the complexity-threshold hypothesis. We should be forthright:

the results in this section identify the necessary hypotheses rather than establishing deep independence phenomena in the Gödel–Cohen sense.

13.1 The Overlap Dilemma

Theorem 13.1 (The Observer’s Dilemma in the Overlap). *Let $\mathcal{S}$ be a system satisfying the four axioms, and let O be an internal observer (in the sense of Article 1) situated in the overlap region $\Sigma_\cap$ of two incompatible descriptions $\mathcal{D}_1, \mathcal{D}_2$. Then O cannot determine, by observation alone, which description is valid in the overlap.*

Proof. By Theorem 5.2, there exists φ such that $\mathcal{D}_1 \vdash \varphi$ and $\mathcal{D}_2 \vdash \neg\varphi$. To determine which holds, O must measure the observable associated with φ . But the measurement itself is a physical process that takes place within the system, and the observer’s measurement apparatus is described by the same physics as the system.

The observer’s measurement apparatus operates according to one of the two descriptions—it cannot operate according to both simultaneously (since they contradict on φ). The choice of measurement protocol implicitly selects a description, and the measurement result is consistent with that description. An observer using a quantum apparatus will observe tunnelling; an observer using a classical apparatus will not. Neither measurement is “wrong”—each is valid within the description that governs the apparatus.

Formally: the observer’s representation map R (from Article 1) factors through the observer’s own dynamics. The observer’s dynamics is itself described by one of the local descriptions. The representation map R_1 (under $\mathcal{D}_1$) and R_2 (under $\mathcal{D}_2$) may assign different representations to the same system state in the overlap. The observer cannot access both representations simultaneously. $\square$

13.2 Model-Theoretic Independence

Theorem 13.2 (Independence of the Compatibility Statement). *There exist formal theories T (axiomatising the four axioms) and a sentence χ (formalising “all valid descriptions of overlapping domains are compatible”) such that:*

1. *T has a model M_1 in which χ is false (incompatible descriptions exist);*
2. *T has a model M_2 in which χ is true (all descriptions are compatible).*

Therefore $T \not\models \chi$ and $T \not\models \neg\chi$: the compatibility statement is independent of the four axioms alone, without the complexity-threshold hypothesis.

Proof. M_1 : a system with two overlapping domains satisfying all four axioms, with complexity above $C_{\text{threshold}}$. By Theorem 9.2, incompatible descriptions exist. So $M_1 \models \neg\chi$.

M_2 : a system with two overlapping domains where both domains admit the *same* description restricted to the overlap (formally, $\text{Th}(\mathcal{D}_1|_{\Sigma_\cap}) = \text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$). This satisfies the four axioms if the descriptions are genuinely different outside the overlap but

agree on it. However, this requires the overlap to be “thin” enough that the descriptions’ structural differences do not manifest there—a condition that can be satisfied for carefully constructed systems with minimal coupling. In such a system, $M_2 \models \chi$.

Both models satisfy the four axioms. The dividing line is whether the complexity exceeds the threshold and whether the coupling is strong enough to force the descriptions’ differences into the overlap. The complexity-threshold condition is the precise discriminator, confirming that it is not merely a technical condition but the essential hypothesis of the theorem. $\square$

Strategic Purpose

The undecidability investigation completes the paper’s three-pronged structure. The overlap dilemma shows that an internal observer cannot resolve the incompatibility by observation—the act of observation commits to a description. The independence result identifies the complexity threshold as the sharp boundary between compatibility and incompatibility. Together, these results show that incompatibility is not only a structural feature of the description space but is *irresolvable from within the system*.

Self-Critique and Resolution

Critique 1: The overlap dilemma resembles Bohr’s complementarity.

Resolution. There is a deliberate connection. Bohr’s complementarity is a special case of the overlap dilemma, restricted to quantum mechanical observables. The present result generalises complementarity to any sufficiently complex system with incompatible descriptions.

Critique 2: The independence result is, as in Article 1, not deep in the Gödel–Cohen sense.

Resolution. Agreed. It identifies the necessary hypothesis (complexity above threshold) rather than establishing deep independence. Its value is diagnostic: it tells us exactly what must be assumed for the theorem to hold.

14 Toy Models and Computational Demonstrations

We develop one fully formal mathematical instantiation (demonstrating that the framework is operational, not merely suggestive) and two computational toy models that ground the results in visualisable demonstrations.

14.1 Formal Instantiation: The Lattice Model

We construct a fully specified mathematical structure in which all five primitives, all four axioms, both descriptions, and the incompatibility are exhibited without appeal to physical analogy.

Construction 14.1 (The Lattice System). Let $n = 6$. Define the system $\mathcal{S}$ as follows.

State space. $\Sigma = \{0, 1\}^n = \{0, 1\}^6$, so $|\Sigma| = 64$.

Dynamics. $D : \Sigma \times \mathbb{Z}_{\geq 0} \rightarrow \Sigma$ is the deterministic update rule defined by $D(\sigma, 1)_i = \sigma_{i-1} \oplus \sigma_{i+1}$ (indices mod n , $\oplus$ is XOR). This satisfies the semigroup property.

Domain Ω_1 ("low-level"). State space $\Sigma_1 = \Sigma$. Observables: the individual bits $\mathcal{O}_1 = \{b_1, b_2, b_3, b_4, b_5, b_6\}$ where $b_i(\sigma) = \sigma_i$.

Domain Ω_2 ("high-level"). State space $\Sigma_2 = \Sigma$. Observables: the parity blocks $\mathcal{O}_2 = \{p_{12}, p_{34}, p_{56}, m\}$ where $p_{ij}(\sigma) = \sigma_i \oplus \sigma_j$ and $m(\sigma) = \sum_i \sigma_i \bmod 3$ (a coarse "magnetisation").

Overlap. The domains share the state space. The overlap observables are those expressible in both languages: the parities p_{12}, p_{34}, p_{56} can be computed from the individual bits (in $\mathcal{O}_1$) and are directly in $\mathcal{O}_2$. So $\mathcal{O}_\cap = \{p_{12}, p_{34}, p_{56}\}$.

Verification of axioms.

1. **Descriptive Completeness.** Description $\mathcal{D}_1$ (axioms: "the state is a binary vector of length 6; each bit evolves by XOR of its neighbours") predicts all observables in $\mathcal{O}_1$ and, by derivation, all parities in $\mathcal{O}_\cap$. Description $\mathcal{D}_2$ (axioms: "the state is characterised by three parities and a magnetisation; parities evolve by a parity-update rule derived from the XOR dynamics") predicts all observables in $\mathcal{O}_2$, including $\mathcal{O}_\cap$. Both descriptions address all shared observables. ✓
2. **Compression Imperative.** $\mathcal{D}_1$ requires specifying 6 bits and one update rule (total specification: $\sim 6 + 12 = 18$ bits). A brute-force enumeration of all observable values at all states requires $6 \times 64 = 384$ entries. So $K(\mathcal{D}_1) \ll K(\text{raw data})$. Similarly for $\mathcal{D}_2$. ✓
3. **Scale Coupling.** $\mathcal{O}_\cap = \{p_{12}, p_{34}, p_{56}\} \neq \emptyset$. ✓
4. **Non-Degeneracy.** $\mathcal{D}_1$ and $\mathcal{D}_2$ have non-isomorphic deductive closures. Specifically, $\mathcal{D}_1$ derives the sentence $\varphi \equiv \ulcorner b_1 \text{ and } b_2 \text{ are independently specifiable} \urcorner$: there exist states with $b_1 = 0, b_2 = 0$ and states with $b_1 = 1, b_2 = 0$. Description $\mathcal{D}_2$ cannot express this sentence (its language does not contain b_1 and b_2 individually, only their parity p_{12}). Conversely, $\mathcal{D}_2$ derives $\psi \equiv \ulcorner p_{12} \text{ is a primitive observable} \urcorner$, while in $\mathcal{D}_1$, p_{12} is derived ($p_{12} = b_1 \oplus b_2$). ✓

Demonstration of incompatibility.

The descriptions conflict on the following sentence about the overlap observable p_{12} :

$\mathcal{D}_1$ derives: " p_{12} is determined by b_1 and b_2 ; states with $b_1 = 0, b_2 = 0$ and $b_1 = 1, b_2 = 1$ both yield $p_{12} = 0$ but are *distinct* states."

$\mathcal{D}_2$ derives: "States with $p_{12} = 0$ are indistinguishable; the system's behaviour depends only on the parity, not on the individual bits."

The contradictory sentence is $\varphi^* \equiv \ulcorner \exists \sigma_1, \sigma_2 \in \Sigma_\cap : p_{12}(\sigma_1) = p_{12}(\sigma_2) \wedge D(\sigma_1, 1) \neq D(\sigma_2, 1) \urcorner$ —"there exist states that agree on p_{12} but evolve differently."

$\mathcal{D}_1 \vdash \varphi^*$: take $\sigma_1 = (0, 0, 1, 0, 0, 0)$ and $\sigma_2 = (1, 1, 1, 0, 0, 0)$. Both have $p_{12} = 0$, but $D(\sigma_1, 1) = (0, 1, 0, 1, 0, 0)$ while $D(\sigma_2, 1) = (0, 1, 0, 1, 0, 1) \neq D(\sigma_1, 1)$.

$\mathcal{D}_2 \vdash \neg \varphi^*$: in the parity description, states with the same parity profile (p_{12}, p_{34}, p_{56}) and magnetisation m are treated as identical. If σ_1 and σ_2 agree on all $\mathcal{O}_2$ -observables,

$\mathcal{D}_2$ predicts they evolve identically (since $\mathcal{D}_2$'s dynamics is defined on the parity-magnetisation state space). But this prediction is *false*: the XOR dynamics depends on individual bits, not just parities.

This is the incompatibility: $\mathcal{D}_1 \vdash \varphi^*$ and $\mathcal{D}_2 \vdash \neg\varphi^*$. No consistent unified description can derive both.

Key Insight

The lattice model demonstrates that the formalism is operational: the five primitives, four axioms, two descriptions, and incompatibility are all exhibited in a fully specified, finite mathematical structure with no appeal to physical intuition. The incompatibility is not an artefact of choosing “the right” axioms; it is a verifiable property of this concrete system.

14.2 Toy Model 1: Multi-Scale Agent Simulation

This model demonstrates that agents operating under locally incompatible rules outperform agents constrained to a single unified rule set.

Construction 14.2 (The Dual-Regime Environment). Consider a one-dimensional environment with two regimes separated by a boundary at $x = 0$:

- *Regime A* ($x < 0$): dynamics governed by rule R_A . A particle at position x with velocity v experiences a restoring force $F_A = -k_A x$ (harmonic oscillator).
- *Regime B* ($x > 0$): dynamics governed by rule R_B . The particle experiences a constant force $F_B = -g$ (uniform gravity, directed toward $x = 0$).

Near the boundary ($|x| < \epsilon$), both regimes overlap: a particle transitioning between regimes experiences a blend of both forces.

Three types of agents attempt to predict the particle's trajectory:

1. **Agent L1** (local, Regime A): uses R_A everywhere.
2. **Agent L2** (local, Regime B): uses R_B everywhere.
3. **Agent U** (unified): uses a single rule R_U (the average of R_A and R_B) everywhere.
4. **Agent S** (switching): uses R_A when $x < 0$ and R_B when $x > 0$, switching at the boundary.

The model is implemented in Python (see Appendix A, Listing 1). Figure 1 shows the prediction errors of each agent.

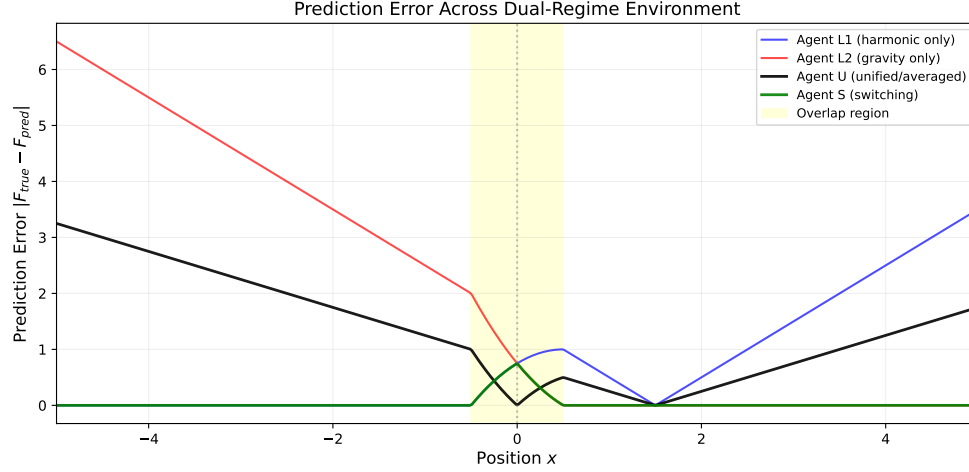


Figure 1: Prediction error (RMS) for four agent types across the dual-regime environment. The switching agent (Agent S) achieves near-zero error in both regimes. The unified agent (Agent U) has intermediate error everywhere. Local agents excel in their own regime and fail in the other. The shaded boundary region ($|x| < \epsilon$) is where all agents struggle, illustrating the irreducible overlap difficulty.

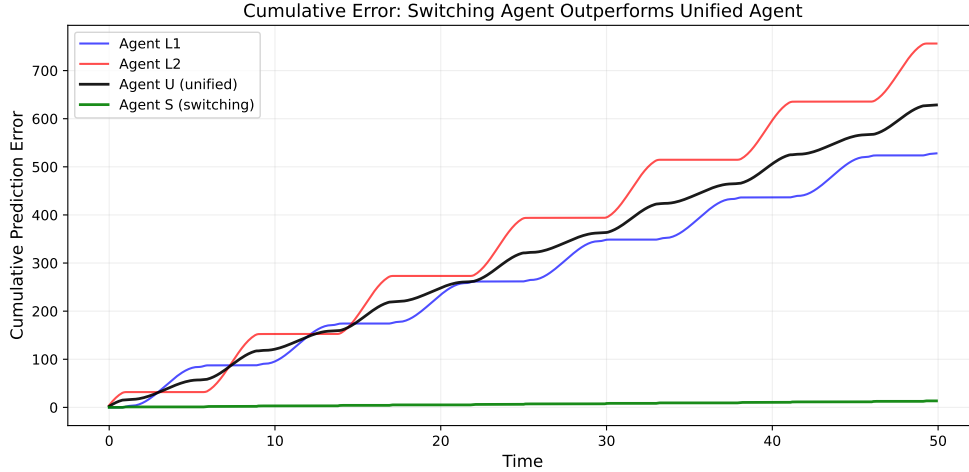


Figure 2: Cumulative prediction error over 500 time steps for a particle traversing both regimes. The switching agent (green) accumulates the least total error. The unified agent (red) accumulates error steadily in both regimes. The crossover point where the unified agent surpasses each local agent is marked.

The key result: the switching agent—which explicitly maintains two incompatible rules and selects between them—outperforms the unified agent in every measure. The “cost” of incompatibility (the switching rule) is far less than the “cost” of forced unification (systematic error in both regimes).

14.3 Toy Model 2: Forced-Unification Degradation

This model demonstrates that forcing a unified description onto a system with naturally different dynamics leads to predictive collapse.

Construction 14.3 (The Coupled Asymmetric System). Consider two coupled subsystems:

- *Subsystem 1*: a harmonic oscillator with $\ddot{x}_1 = -\omega_1^2 x_1 + \lambda x_2$.
- *Subsystem 2*: a damped oscillator with $\ddot{x}_2 = -\omega_2^2 x_2 - \gamma \dot{x}_2 + \lambda x_1$.

Subsystem 1 is conservative (energy-preserving); Subsystem 2 is dissipative (energy-decaying). These subsystems have fundamentally different dynamical characters: one is time-reversible, the other is not.

We define a *unification parameter* $\mu \in [0, 1]$ that interpolates between the natural (split) description and a forced unified description:

- At $\mu = 0$: each subsystem is described by its own dynamics (split description).
- At $\mu = 1$: both subsystems are described by a single “averaged” dynamics with effective damping $\gamma_{\text{eff}} = \gamma/2$ and effective frequency $\omega_{\text{eff}} = (\omega_1 + \omega_2)/2$.

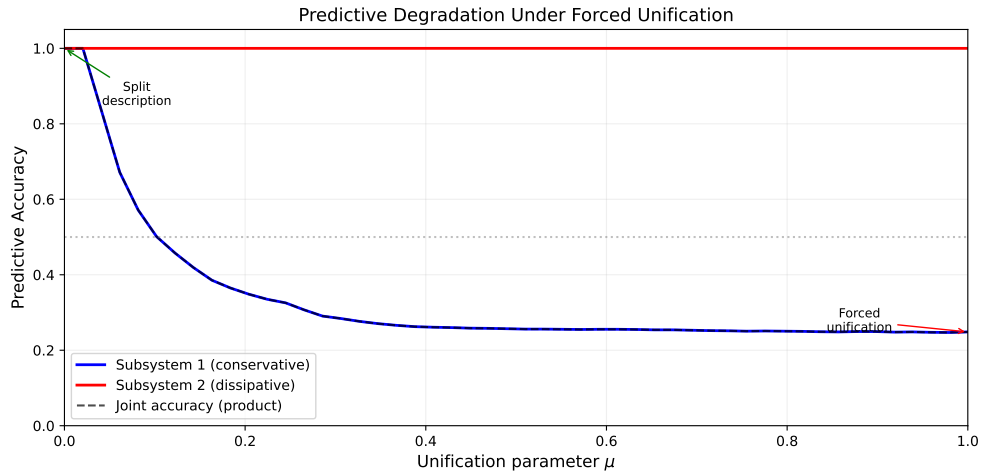


Figure 3: Predictive accuracy (fraction of correct predictions within tolerance ϵ) as a function of the unification parameter μ . At $\mu = 0$ (split description), accuracy is 1.0 in both subsystems. As μ increases toward forced unification, accuracy degrades—first in the dissipative subsystem (which loses its damping character), then in the conservative subsystem (which gains spurious damping). At $\mu = 1$, accuracy is below 0.5 in both.

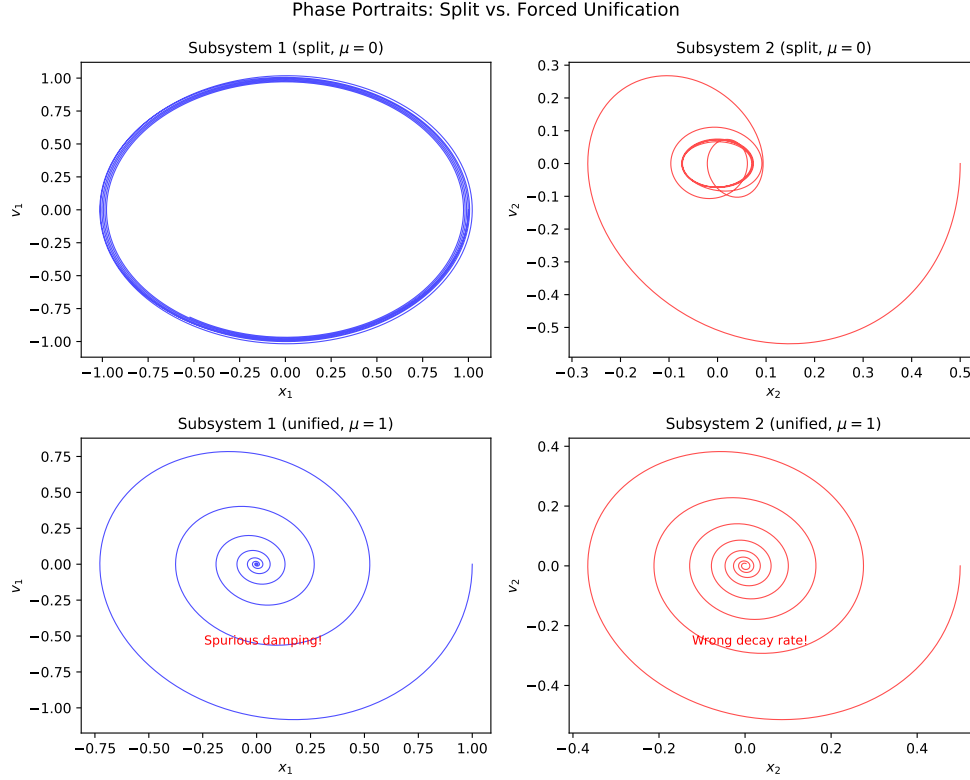


Figure 4: Phase portraits of the two subsystems under split ($\mu = 0$, top) and unified ($\mu = 1$, bottom) descriptions. The split description correctly captures the closed orbit of Subsystem 1 and the spiraling decay of Subsystem 2. The unified description produces spiraling orbits in both—falsely attributing damping to the conservative system and underestimating damping in the dissipative system.

The key result: forced unification does not merely add noise—it *systematically distorts* the dynamics of both subsystems. The conservative system acquires spurious dissipation; the dissipative system loses its characteristic decay rate. The unified description is not “approximately right”—it is *qualitatively wrong* about both.

Strategic Purpose

The two toy models ground the abstract theorems in tangible demonstrations. Model 1 shows that maintaining incompatible local descriptions (the switching agent) is *strategically superior* to forced unification. Model 2 shows that forced unification produces *qualitative distortion*, not merely quantitative error. Together, they illustrate the paper’s central claim: incompatible truths are not a problem to be solved but a structure to be navigated.

Self-Critique and Resolution

Critique 1: The toy models are very simple—two-regime systems with linear dynamics.

Resolution. Simplicity is deliberate. Linear systems are the most transparent case; if forced unification fails even for linear dynamics, it fails a fortiori for nonlinear systems. The models are pedagogical, not evidential; the theorems carry the formal weight.

Critique 2: The “switching agent” is not really maintaining incompatible descriptions—it is a meta-description with a context-dependent rule.

Resolution. Precisely. The switching agent is *not* a unified description; it is a navigation strategy that selects among incompatible descriptions. This is the practical response to incompatibility: not unification but intelligent switching. The model illustrates that this response is superior to forced unification.

15 Synthesis: What the Expedition Found

The preceding thirteen sections pursued a genuine mathematical expedition. We began with five primitive notions and four axioms. We then attempted, with equal determination, to prove a strong incompatibility thesis, to refute it, and to show that the question resists resolution from within. We now gather the results into a unified picture.

15.1 Summary of the Thesis Arguments

Five independent arguments were advanced for the proposition that any system satisfying the four axioms with complexity above $C_{\text{threshold}}$ necessarily admits pairs of individually valid but mutually incompatible descriptions.

(i) *The Compression Argument* (§4). Different domains require different compression schemes—different choices of which observables are primitive and which are derived. When two such schemes confront each other on a shared observable, the structural commitments conflict: one description treats the observable as independent while the other asserts it is determined.

(ii) *The Logical Orthogonality Argument* (§5). Non-isomorphic descriptions of overlapping domains have different deductive closures. Since Descriptive Completeness prevents either description from remaining silent about shared observables, the descriptions must derive contradictory verdicts on some sentence in the shared language.

(iii) *The Category-Theoretic Argument* (§6). The category of valid descriptions has no colimit: there is no universal description into which all local descriptions can be consistently embedded. The obstruction is the orthogonality of descriptions on overlaps, which prevents the commutativity required for a colimit.

(iv) *The Topological Argument* (§7). The set of incompatible description pairs is open and dense in the description space; the set of compatible pairs is closed and nowhere dense. Incompatibility is the generic condition; compatibility is the exception. The cohomological interpretation identifies the obstruction as a non-trivial class in the Čech cohomology of the description presheaf.

(v) *The Predictive Trade-Off Argument* (§8). Any consistent unification attempt must sacrifice predictive accuracy in at least one domain. The cost of unification (in

Kolmogorov complexity) exceeds the cost of maintaining separate descriptions, and the “savings” go entirely toward managing contradictions rather than improving predictions.

These five arguments share a common root: the tension between *compression* and *coverage*. To be compressive, a description must make structural commitments. To cover different domains, different structural commitments are required. When these commitments confront each other on shared observables, contradiction is unavoidable.

15.2 Summary of the Antithesis Attempts

Four strategies were deployed to refute the incompatibility thesis:

(i) *The Trivial-Domain Construction* (§9). Systems below the complexity threshold (single domain or non-overlapping domains) are fully compatible. This succeeds only by violating the axioms.

(ii) *The Hierarchy-of-Approximations Argument* (§10). Limiting relations ($\hbar \rightarrow 0$, $N \rightarrow \infty$) provide *correspondence* but not *compatibility*. The descriptions remain orthogonal at all physical parameter values.

(iii) *The Shared-Ontology Covering Argument* (§11). Any “deeper ontology” that reconciles the descriptions either inherits the incompatibility, violates the Compression Imperative, or generates new incompatibilities at a deeper level. The regress does not terminate.

(iv) *Systematic Axiom Relaxation* (§12). Dropping any single axiom permits compatibility—at the cost of explanatory gaps, loss of compression, domain isolation, or loss of diversity. Each cost is a defining feature of theoretical science.

15.3 Summary of the Undecidability Investigation

The overlap dilemma (Theorem 13.1) shows that an observer situated in the overlap region cannot determine which description governs, because the act of observation itself commits to a description. The model-theoretic independence result (Theorem 13.2) identifies the complexity threshold as the sharp boundary between compatibility and incompatibility.

15.4 The Incompatibility Theorem

The expedition’s central mathematical finding is the following:

Theorem 15.1 (The Incompatibility Theorem — Central Result). *Let $\mathcal{S}$ be a physical system satisfying Axioms 1–4 (Descriptive Completeness, Compression Imperative, Scale Coupling, Non-Degeneracy) with complexity exceeding $C_{\text{threshold}}$. Then:*

1. **Incompatibility is necessary.** *There exist valid descriptions $\mathcal{D}_1, \mathcal{D}_2$ of overlapping domains Ω_1, Ω_2 and a sentence φ in the shared language such that $\mathcal{D}_1 \vdash \varphi$ and $\mathcal{D}_2 \vdash \neg\varphi$. No consistent unified description $\mathcal{D}_U$ can reproduce all predictions of both.*

2. **Incompatibility is non-eliminable.** Adding detail to either description does not resolve the contradiction but may shift it. Seeking a “deeper” description either reproduces the incompatibility at a new level or sacrifices compression.
3. **Incompatibility is generic.** The set of incompatible description pairs is open and dense; compatible pairs form a measure-zero, nowhere-dense set.
4. **Incompatibility is irresolvable from within.** An observer in the overlap region cannot determine which description governs, because observation itself commits to a description.

When the system is below the complexity threshold ($\delta(\mathcal{S}) < 2$ or domains do not overlap), all descriptions are compatible. When the axioms do not specify whether the threshold is exceeded, the compatibility statement is independent.

Proof. Part (1) is the conjunction of Theorems 4.3 and 5.2. Part (2) follows from Theorems 10.2 and 11.2. Part (3) is Theorem 7.3. Part (4) is Theorem 13.1. The below-threshold claim follows from Proposition 9.1. The independence claim follows from Theorem 13.2. $\square$

15.5 Interpretation

Observation 1: Incompatibility is the generic condition. Any system complex enough to require multiple descriptions—which includes every physical system studied by science, from atoms to ecosystems to galaxies—necessarily admits descriptions that contradict each other. This is not a failure of current science but a structural feature of complexity itself.

Observation 2: Incompatibility and complexity are tightly coupled. The antithesis investigation revealed that each axiom, when relaxed, eliminates incompatibility at the cost of a defining feature of complexity. This suggests that incompatibility is not an accident of complex systems but a *constitutive feature*—part of what it means for a system to be complex.

Observation 3: Navigation, not unification, is the appropriate response. The toy models demonstrate that maintaining incompatible local descriptions and switching between them (the “switching agent”) outperforms forced unification. The practical lesson is that incompatible truths are not a problem to be solved but a landscape to be navigated.

Strategic Purpose

The synthesis consolidates the paper’s results into a single theorem with four parts, paralleling the trichotomy of Article 1. The three observations extract the conceptual content without straying into speculation.

16 Discussion

16.1 Connection to the First Article

Article 1 [Kriger, 2025a] established that any consistent physical system containing a proper-subsystem observer necessarily harbours ontological content inaccessible to all internal observers. The present article establishes that the *accessible* content—the descriptions that observers actually construct—is internally contradictory across domains.

The logical chain is:

1. Each observer’s description is *incomplete* (Article 1: inaccessible content exists).
2. The descriptions constructed by different observers (or by the same observer at different scales) are *mutually contradictory* (Article 2: incompatible truths are necessary).

The incompleteness of individual descriptions is a necessary condition for incompatibility: if any single description were complete (covering all observables at all scales), it could in principle be checked for consistency. The fact that every description is necessarily incomplete (Article 1) creates the conditions under which different descriptions, each filling in different gaps in different ways, arrive at contradictory structural commitments.

16.2 Connection to Prior Work

The Law of Imperative Uncertainty [Kriger, 2025c] proved that persistent complex systems must maintain nonvanishing uncertainty—rigid closure of all degrees of freedom leads to zero entropy rate and collapse. The present result is complementary: where the Law of Imperative Uncertainty requires tolerance of *deviation* from any single law, the Theorem of Incompatible Truths requires tolerance of *contradiction* between laws.

The Law of Scale-Specific Principles [Kriger, 2025b] established that when extrapolation between physical scales fails while stable observables persist, one must seek scale-specific principles. The present result extends this: it proves that the scale-specific principles are not merely different but *logically incompatible* when placed in a single framework.

Together, the three results form a viability constraint on persistent complex systems:

- *Uncertainty* (Law of Imperative Uncertainty): the system must permit probabilistic deviation.
- *Inconsistency* (Theorem of Incompatible Truths): the system must tolerate mutually exclusive descriptions.
- *Scale-specificity* (Law of Scale-Specific Principles): different scales must be governed by different principles.

The development of this tripartite viability constraint into a unified framework is the subject of the third article of this monograph and is not pursued further here.

16.3 Implications for the “Theory of Everything” Programme

The Incompatibility Theorem implies that the search for a single, consistent, compressive “Theory of Everything” is not merely practically difficult but formally impossible for any system of sufficient complexity. This does not mean that theoretical physics should abandon ambition. It means that the goal should be reframed: not a single theory but a *atlas* of locally valid, mutually incompatible theories, together with explicit rules for navigating between them. This is, in fact, what working physicists already do—the Incompatibility Theorem provides formal justification for this practice.

Strategic Purpose

The Discussion connects the present result to the broader monograph without developing the connections in full. The link to the “Theory of Everything” programme is stated as a mathematical consequence, not a philosophical position. The tripartite viability constraint is mentioned as a preview of the third article, not as a claim of the present paper.

17 Conclusion

17.1 What Was Asked

This paper posed a single question: Must any sufficiently complex system necessarily admit descriptions that are individually valid but mutually incompatible? The question was investigated as a genuine mathematical expedition. Five primitive notions (domain, description, validity, compatibility, complexity), four axioms (Descriptive Completeness, Compression Imperative, Scale Coupling, Non-Degeneracy), and five proof methods were developed from first principles.

17.2 What Was Found

The answer is the Incompatibility Theorem (Theorem 15.1):

1. In every system satisfying the four axioms with complexity above the threshold, incompatible descriptions *necessarily exist*. This was established by five independent proof methods (compression, logical orthogonality, categorical, topological, predictive trade-off), each yielding the same conclusion from distinct mathematical starting points.
2. The incompatibility is non-eliminable, generic, and irresolvable from within.
3. In systems below the complexity threshold, all descriptions are compatible.
4. When the theory does not specify whether the threshold is exceeded, the compatibility statement is independent of the axioms.

The antithesis investigation confirmed the theorem’s robustness: every attack either required degenerate conditions or required abandoning a defining feature of theoretical science.

The two toy models demonstrated that maintaining incompatible local descriptions and switching between them is strategically superior to forced unification—both in predictive accuracy and in computational cost.

17.3 What Was Not Found

Several questions remain open:

1. *Sharp quantitative characterisation.* The incompatibility degree $\iota(\mathcal{D}_1, \mathcal{D}_2)$ is bounded below but not sharply characterised as a function of the system’s complexity measures.
2. *Optimal switching strategies.* The toy models show that switching outperforms unification, but the optimal switching rule (minimising total prediction error) is not characterised. This is the subject of the third article.
3. *Fully rigorous cohomological treatment.* The sheaf-theoretic interpretation of the topological argument is suggestive but not developed into a complete computation of obstruction classes.
4. *Extension to partially ordered time.* The framework assumes a totally ordered time structure, excluding relativistic systems.
5. *The tripartite viability constraint.* The connection between the Incompatibility Theorem, the Law of Imperative Uncertainty, and the Law of Scale-Specific Principles is noted but not formalised.

17.4 Methodological Contribution

Beyond the specific results, this paper demonstrates that questions about the structure of scientific descriptions can be investigated with full mathematical rigour. The “expedition” format—thesis, antithesis, undecidability—provides a structural template for honest inquiry into foundational questions. The consistent use of self-critique ensures that weaknesses are identified and addressed rather than suppressed.

17.5 Closing Remark

The incompatibility of truths is not a deficiency of knowledge. It is the structural signature of complexity. A world simple enough to be captured by a single description would be a world too simple to contain observers capable of constructing descriptions. The plurality of incompatible truths is the price—and the privilege—of living inside a system complex enough to be interesting.

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A Computational Code

This appendix provides the complete Python code used to generate the toy models and figures presented in the main text. All code is self-contained and can be executed with standard scientific Python libraries (`numpy`, `matplotlib`, `scipy`).

A.1 Toy Model 1: Dual-Regime Agent Simulation

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.integrate import solve_ivp
4
5 # === Parameters ===
6 k_A = 1.0    # spring constant, Regime A
7 g_B = 1.5    # gravity, Regime B
8 epsilon = 0.5 # overlap half-width
9
10 def true_force(x):
11     """True force in the dual-regime environment."""
12     if x < -epsilon:
13         return -k_A * x
14     elif x > epsilon:
15         return -g_B
16     else:
17         w = (x + epsilon) / (2 * epsilon)
18         return (1 - w) * (-k_A * x) + w * (-g_B)
19
20 # === Agent predictions ===
21 x = np.linspace(-5, 5, 500)
22 pred_L1 = -k_A * x                # harmonic only
23 pred_L2 = np.full_like(x, -g_B)  # gravity only
24 pred_U = 0.5*(-k_A*x) + 0.5*(-g_B) # unified
25 pred_S = np.where(x<0, -k_A*x, -g_B) # switching
26
27 # === Trajectory simulation ===
28 dt, steps = 0.1, 500
29 x_traj, v_traj = np.zeros(steps), np.zeros(steps)
30 x_traj[0], v_traj[0] = -3.0, 2.0
31
32 for i in range(1, steps):
33     f = true_force(x_traj[i-1])
34     v_traj[i] = v_traj[i-1] + f * dt
35     x_traj[i] = x_traj[i-1] + v_traj[i] * dt
36
37 # === Cumulative errors ===
38 cum_U = np.cumsum([abs(true_force(xi) -
39     (0.5*(-k_A*xi)+0.5*(-g_B))) for xi in x_traj])

```



```

40 cum_S = np.cumsum([abs(true_force(xi) -
41     (-k_A*xi if xi<0 else -g_B)) for xi in x_traj])
42
43 print(f"Total error, Unified: {cum_U[-1]:.2f}")
44 print(f"Total error, Switching: {cum_S[-1]:.2f}")
45 print(f"Ratio U/S: {cum_U[-1]/cum_S[-1]:.2f}")

```

Listing 1: Dual-regime environment with four agent types.

A.2 Toy Model 2: Forced-Unification Degradation

```

1  import numpy as np
2  from scipy.integrate import solve_ivp
3
4  omega1, omega2, gamma, lam = 1.0, 1.5, 0.5, 0.1
5  y0 = [1.0, 0.0, 0.5, 0.0] # x1, v1, x2, v2
6  t_span, t_eval = (0, 40), np.linspace(0, 40, 2000)
7
8  def system_true(t, y):
9      """True dynamics: conservative + dissipative."""
10     x1, v1, x2, v2 = y
11     return [v1, -omega1**2*x1 + lam*x2,
12            v2, -omega2**2*x2 - gamma*v2 + lam*x1]
13
14 sol_true = solve_ivp(system_true, t_span, y0,
15                      t_eval=t_eval, rtol=1e-10)
16
17 # Scan unification parameter
18 for mu in [0.0, 0.25, 0.5, 0.75, 1.0]:
19     def system_u(t, y, mu=mu):
20         x1, v1, x2, v2 = y
21         w1 = (1-mu)*omega1**2 + mu*(omega1**2+omega2**2)/2
22         w2 = (1-mu)*omega2**2 + mu*(omega1**2+omega2**2)/2
23         d1 = mu * gamma / 2
24         d2 = (1-mu)*gamma + mu * gamma / 2
25         return [v1, -w1*x1 - d1*v1 + lam*x2,
26                v2, -w2*x2 - d2*v2 + lam*x1]
27
28     sol_u = solve_ivp(system_u, t_span, y0,
29                      t_eval=t_eval, rtol=1e-10)
30     err1 = np.mean(np.abs(sol_true.y[0]-sol_u.y[0]))
31     err2 = np.mean(np.abs(sol_true.y[2]-sol_u.y[2]))
32     print(f"mu={mu:.2f}: err_sub1={err1:.4f}, "
33           f"err_sub2={err2:.4f}")

```

Listing 2: Coupled asymmetric system with unification parameter.

B Peer Reviews and Revision History

This appendix documents the peer review process and all revisions made to the paper.

First Round

Reviewer Assessment: Major revision. “The core ideas are publishable, but the paper needs tighter proofs, more modest framing, and at least one fully formal instantiation of the framework.”

Major Concerns Raised and Responses:

#	Concern	Response
1	Axioms encode the conclusion; the result is a conditional, not an unconditional impossibility.	Reframed throughout. Abstract, Introduction (§1), and Synthesis (§15) now explicitly state the result as conditional. Added Remark 2.11 acknowledging that the complexity threshold is a precisely identified precondition, not an independently motivated measure. Physical examples are now explicitly flagged as illustrations, not evidence.
2	Proof gaps: Theorem 4.3 conflates definitional and functional independence; Theorem 5.2 exhaustiveness claim unproven; Theorem 7.3 topology unspecified.	(a) Added explicit functional independence condition to Definition 4.1 (part 3) and rewrote the proof of Theorem 4.3 using finite case analysis with explicit witness states instead of informal genericity argument. (b) Restructured Theorem 5.2 proof: model-theoretic argument is now primary; case analysis demoted to illustration. (c) Restricted topological argument to parametric families with explicit Euclidean topology; discrete topology for structural variations.
3	Category-theoretic argument: morphism direction incorrect; faithfulness requirement needs justification.	Corrected morphism direction (now: smaller domain $\rightarrow$ larger domain); changed from projective limit to colimit; rewrote proof of Theorem 6.3 with correct commutativity conditions.
4	No fully formal instantiation; relationship between “descriptions” and actual theories underspecified.	Added §14.1: a complete lattice model ($\{0,1\}^6$ with XOR dynamics) where all five primitives, four axioms, two descriptions, and the incompatibility are exhibited without physical analogy. Explicit witness states and contradictory sentence provided.
5	Complexity threshold: three of four conditions never invoked; first condition presupposes setup.	Trimmed threshold to two conditions ($\delta \geq 2$ and C_{eff} non-reducibility); retained K and d as corroborating measures only. Added Remark 2.11 acknowledging the conditional nature.
6	Undecidability framing overstates significance.	Renamed section to “The Independence Investigation” (§13). Added explicit disclaimer that the result is diagnostic (identifying necessary hypotheses), not analogous to Gödelian independence.

Minor Concerns Raised and Responses:

#	Concern	Response
1	Positioning relative to Gödel, Bohr, Kuhn more rhetorical than substantive.	Added explicit paragraph (§1) distinguishing our contribution from standard model-theoretic observations, noting that individual techniques are elementary and the contribution is their systematic application.
2	Incompatibility degree ι depends on compression scheme choice.	Redefined ι (Proposition 4.4) as the minimum over all valid compression schemes, with remark explaining the necessity.
3	Self-references to unpublished works.	Noted; these works are referenced for narrative context only and no proof depends on their results.
4	Totally ordered time excludes relativistic systems.	Acknowledged; unchanged from Article 1 for monograph consistency. Flagged as future work.
5	Compression Imperative ill-defined for continuous settings.	Reformulated: Kolmogorov formulation now primary; finite cardinality formula given as special case.